

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC Engineering Services

Subject: Engineering

Questions: 367

Source: Official source PDF extraction

AI-generated PYQs: 0

1. [UNKNOWN] unknown

0) proportional-plus-integral control proportional-plus-derivative control proportional-plus-integral-plus- derivative control

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2. [UNKNOWN] unknown

1. Which of the following statements are wholly correct regarding broken-brick aggregate useable in concretes?

Broken-brick aggregate is obtained by crushing waste bricks; and it has a density varying between 1000 kg/m³ — 1200 kg/m³. Such aggregate is usable in concrete for foundation in light buildings, floorings and walkways. Such aggregate may also be used in light-weight reinforced concrete floors. 1 and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3

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3. [UNKNOWN] unknown

1. Not using vibrator to spread the concrete Reducing the continued vibration Improving the cohesion of a lean dry mix through addition of a further small quantity of water 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only

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4. [UNKNOWN] unknown

1. Its design corresponds to soft sandy foundation. The difference in weir crest and downstream river bed may not exceed

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5. [UNKNOWN] unknown

1. Which one of the following is not a component of 'Capital Receipts'? Market borrowings including special bonds External loans raised by the Central Government from abroad Receipts from taxes on property and capital transactions Provident Funds (State 'Provident Funds and Public Provident Fund)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

6. [UNKNOWN] unknown

1. Water is discharged from a tank 4. A nozzle at the end of an 80 mm hosepipe maintained at a constant head of 5 m produces a jet 40 mm in diameter. above the exit of a straight pipe 100 m When it is discharging the water long and 15 cm in diameter. If the 1200 Lpm, the force on the joint at the friction coefficient for the pipe is 0.01, base of the nozzle will be the rate of flow will be nearly 0.04 m³ /s 005 M³ /s 0.06 m³ /s 0.07 m³ /s 180 N 200 N 220 N 240 N

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7. [UNKNOWN] unknown

2. In handling air-entraining admixtures the beneficial amount of entrained air depends upon certain factors like Type and quantity of air-entraining agent Water-cement ratio of the mix Strength of aggregates Extent of compaction of concrete 1, 2 and" only 1, 2 and 4 only 1, 3 and 4 only 1, 2, 3 and 4

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8. [UNKNOWN] unknown

2. Which one of the following statement is correct with respect to the 'societal development'? Behaviour grows into habits, habits into tradition and tradition becomes custom Customs grow into mores and mores grow into custom Behaviours grow into customs and customs grow into traditions Folkways grow into tradition and traditions grow into customs

9. [UNKNOWN] unknown

2. A plate weighing 150 N and measuring 0.8 m x 0.8 m just slides down an inclined plane over an oil film of 2 mm thickness for an inclination of 30° and velocity of 0.2 m/s. Then the viscosity of the oil used is

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

10. [UNKNOWN] unknown

3. Which one of the following statements is not correct with respect to fly ash? As part replacement of cement in the range of 15% — 30%, fly ash reduces the strength in the initial period, but once the Pozzolanic process sets in, higher strength can be obtained. Fly ash as a part replacement of sand has a beneficial effect on strength even at early age. Fly ash as a part replacement of sand is economical. A simultaneous replacement of cement and fine aggregates enables the strength at a specified age to be equalled depending upon the water content.

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11. [UNKNOWN] unknown

3. When water passes over it, the longitudinal location of the formation of a hydraulic jump is variable. This weir is of the type Vertical drop weir Masonry or concrete sloping weir Dry stone slope weir. Parabolic weir.

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12. [UNKNOWN] unknown

3. Which one of the following statement is correct with respect to 'the convergence theory' on social change? The societal adaptive culture is changing more slowly As societies become modernized, they begin to resemble one another more closely The developed countries show more growth in social changes than the less developed countries Strong opposition by old people retards the social change

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13. [UNKNOWN] unknown

3. A spherical balloon of 1.5 m diameter is completely immersed in water and chained to the bottom. If the chain has a tension of 10 kN, the weight of the balloon will be nearly

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14. [UNKNOWN] unknown

4. Which one of the following statements is not correct with respect to the properties of cement? Highly reactive Pozzolanas enhance the early age strength of the composite cement. Pozzolanic activity refines pore structure which decreases electrolytic resistance of concrete. The expansion due to alkali-silica reaction can be controlled by replacement of as high as 60% of OPC with high-calcium Pozzolana. Such high amounts of replacement cements result in higher accelerated carbonation depths compared to pure use of OPC only.

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15. [UNKNOWN] unknown

4. Draft tube 1, 2 and 3 only 1, 2 and 4 only 1, 3 and 4 only 2, 3 and 4 only

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16. [UNKNOWN] unknown

4. With respect to the conduct and performance of a company, 'perfect competition' refers to Large number of small firms producing differentiated products Complete freedom in economic life and absence of rivalry among firms Many companies selling similar products with free entry Sole producer selling a distinct product

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17. [UNKNOWN] unknown

4. Speed of train is $\frac{1}{2}$ if $\frac{1}{2} = 5 \times 10$ Select the correct answer using the codes given below: 1 and 4 only 2 and 4 only

1 and 3 only 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

18. [UNKNOWN] unknown

5. Hydration of which compound is responsible for increase in strength of cement in later age? Tr-calcium Aluminate (C3A) Tetra-calcium Alummoferrite (C4AF) Tr-calcium Silicate (C3S) Di-calcium Silicate (C2S)

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19. [UNKNOWN] unknown

5. The cheapest method of marketing of securities with the only cost incurred being on sending 'letters of rights' to existing holders is Public issue through prospectus method Offer for sale method Rights issue Subscription by inside coterie method

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20. [UNKNOWN] unknown

5. A vertical water pipe, 1.5 m long, tapers from 75 mm diameter at the bottom to 150 mm diameter at the top and the rate of flow is 50 L/s upwards. If the pressure at the bottom end is 150 kN/m², the pressure at the top will be nearly (a) 0.3 N s/m² (a) 1952 . kN/m² (b) 0.4 N s/m² 1914 kN/m² 0.5 N s/m² (c) 187.6 kN/m² (d) 0.7 N s/m² (d) 183.8 kN/m²

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21. [UNKNOWN] unknown

6. The creep strain of cement attains its terminal value by 1 year 2 years 5 years 6 months VBY-S-CVL (2 — A) =====
PAGE 3 =====

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22. [UNKNOWN] unknown

6. 'Fiscal policy' means Balancing the revenue collection and expenditure Establishing equilibrium between demand and supply of goods and services Use of taxation, public borrowing and public expenditure by Govern- ment for purposes of 'stabilization' or 'development' Deficiency as an instrument of growth V13Y-G-DST — A 2 ===== PAGE 3 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

23. [UNKNOWN] unknown

6. The stream function for a flow field is $w = 3x^2y + (2 + t)y^2$. The velocity at a point P for position vector $r = i + 2j - 3k$ at time $t = 2$ will be (a) 9.11 kN (a) $19i - 12j$ (b) 8.22 kN $21i - 12j$ 644 kN (c) $19i + 22j$ (d) 7.33 kN (d) $21i + 22j$ VBY-S-MCH/ 50A 2 ===== PAGE 3 ===== an incompressible flow an unsteady and incompressible flow an unsteady and compressible flow 8300 m 8240 m 8110 m 8000 m

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24. [UNKNOWN] unknown

7. Which of the following methods will help in reducing segregation in concrete?

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25. [UNKNOWN] unknown

7. Which of the following come under the offerings of 'MUDRA' Bank? Portfolio Credit Guarantee Credit for large industries MUDRA Card Credit Enhancement Select the correct answer using the codes given below: 1, 2 and 3.c:only 1, 3 and 4 only 1, 2 and 4 only 2, 3 and 4 only

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26. [UNKNOWN] unknown

7. In a laminar flow through pipe, the point of maximum instability exists at a distance of y from the wall which is of pipe radius R of pipe radius R of pipe radius R of pipe radius R ,,

27. [UNKNOWN] unknown

8. On an average, in a 125 mm slump, the concrete may lose about (in first one hour) 15 mm of slump 25 mm of slump 40 mm of slump 50 mm of slump

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28. [UNKNOWN] unknown

8. Which of the following is/are the key reasons for encouraging start up Entrepreneurship? Innovations Focusing on service industry Bringing the values of proactivity into the society Select the correct answer using the codes given below:

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29. [UNKNOWN] unknown

8. $v = -u$ for a turbulent flow ax signifies conservation of bulk momentum transport increase in u in x -direction followed by increase in it in negative y -direction turbulence is anisotropic turbulence is isotropic

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30. [UNKNOWN] unknown

9. Permeability in concrete is studied towards providing for, or guarding against, which of the following features? The penetration by materials in solution may adversely affect the durability of concrete; moreover, aggressive liquids 'attack' the concrete. In case of reinforced concrete, ingress of moisture and air will result in corrosion of steel leading to an increase in volume of steel, resulting in cracking and swelling of the concrete cover. The moisture penetration depends on permeability and if the concrete can become saturated with water it is less vulnerable to frost action. (a) 1, 2 and 3 (h) 1 and 2 only 1 and 3 only 2 and 3 only (2 iaL) — $(1 - 2p)(1 + p)$ where V is pulse velocity, in mm/s, n is resonant frequency of longitudinal vibration, in Hz, L is distance between transducers, in mm.

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31. [UNKNOWN] unknown

9. Which of the following are the main objectives of Gold Monetization Scheme launched in the country? To monetize gold holdings in the country To increase export of gold from the country To reduce India's import bill To meet the targets of reduction in fiscal deficit Select the correct answer using the codes given below: 1 and 4 only 2 and 4 only 2 and 3 only 1 and 3 only

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32. [UNKNOWN] unknown

9. A flow field satisfying $\nabla \cdot \mathbf{V} = 0$ as the continuity equation represents always (a) a steady compressible flow

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33. [UNKNOWN] unknown

10. Poisson's ratio of concrete μ can be determined using the formula $(2nL(1 - 2p)(1 - \mu))$ (a) $(1 - p)2nL(1 - 21..0(1 + \frac{472}{j} (1 - \mu^2))2nL(1 - 2p)(1 +$

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34. [UNKNOWN] unknown

10. A person travelled by car 70 km towards north to A then covered 30 km turning left to B. Again he turned towards left and travelled 110 km to C. Then he cycled at the rate of 10 km/hour towards the starting point. The time taken by him to reach the starting point from C will be (a) 1 only (a) 3 hours (b) 2 only 5 hours 1 and 3 only (c) 7 hours (d) 1, 2 and 3 (d) 21 hours 3 A — VBY-G-DST ===== PAGE 4 ===== $2r^2(n + 24) r^2(n + 24) 2r^2Or + 36) (d) r^2(it + 36)$

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35. [UNKNOWN] unknown

10. An oil of viscosity 8 poise flows between two parallel fixed plates, which are kept at a distance of 30 mm apart. If the drop of pressure for a length of 1 m is $0.3 \times 10^4 \text{ N/m}^2$ and width of the plates is 500 mm, the rate of oil flow between the plates will be $4.2 \times 10^{-3} \text{ m}^3/\text{s}$ $54 \times 10^{-3} \text{ m}^3/\text{s}$ $6.6 \times 10^{-3} \text{ m}^3/\text{s}$ $7.8 \times 10^{-3} \text{ m}^3/\text{s}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

36. [UNKNOWN] unknown

11. Which one of the following methods/techniques will be used for placing of concrete 'in dewatered 'Caissons or Coffers' dams? Tremie method Placing in bags Prepacked concrete In-the-dry practice

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

37. [UNKNOWN] unknown

11. A student purchases some books for Rs. 1600. If he had bought 8 more books for the same amount, each book would cost Rs. 10 less. The number of books he buys is 30 32 34 36

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38. [UNKNOWN] unknown

11. In case of transmission of hydraulic power by a pipeline to a turbine in a hydroelectric power station, the maximum power transmission efficiency through the pipeline is 76% 67% 54% 42%

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39. [UNKNOWN] unknown

12. The minimum cement content (kg/m^3) for a pre-specified strength of concrete (using standard notations) premised on 'free water-cement ratio' will be as (a) 1, 1000 So 1000 Water Content Water Cement ratio Water Content x Water Cement ratio $100 F C + F V (1 + p)$ (b) $V B Y - S - C V L (3 - A)$ ===== PAGE 4 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

40. [UNKNOWN] unknown

12. A hemisphere depression is cut out from one face of the cubical wooden block such that the radius r of the hemisphere is equal to half of the edge of the cube. What will be the surface area of the remaining solid?

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41. [UNKNOWN] unknown

12. A pipe, having a length 200 m and 200 mm diameter with friction factor 0.015, is to be replaced by a 400 mm diameter pipe of friction factor 0.012 to convey the same quantity of flow. The equivalent length of the new pipe for the same head loss will be $V B Y - S - M C H / 50 A 3$ [P.T.O. ===== PAGE 4 =====

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42. [UNKNOWN] unknown

13. A bar specimen of 36 mm diameter is subjected to a pull of 90 kN during a tension test. The extension on a gauge length of 200 mm is measured to be 0.089 mm and the change in diameter to be 0.0046 mm. The Poisson's ratio will be 0.287 0.265 0.253 0.241

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43. [UNKNOWN] unknown

13. A rod of length 1 is to be divided into two parts, such that if 5 times the smaller portion is added to half of the larger portion, it will always be less than 1. This can be achieved by taking length of the larger portion more than (d)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

44. [UNKNOWN] unknown

13. Certain quantities cannot be located on the graph by a point but are given by the area under the curve corresponding to the process. These quantities in concepts of thermodynamics are called as cyclic functions point

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45. [UNKNOWN] unknown

14. A steel rod 15 m long is at a temperature of 15°C. The values of $\alpha = 12 \times 10^{-6} \text{ m/m}^\circ\text{C}$ and $E = 200 \text{ GN/m}^2$ are adopted. When the temperature is raised to 65°C, what is the free expansion of the length; and if this expansion of the rod is fully prevented, what is the temperature stress produced? 5 mm and 120 MN/m² 9 mm and 120 MN/m² 5 mm and 150 MN/m² 9 mm and 150 MN/m²

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46. [UNKNOWN] unknown

14. Which of the following conditions hold good for a train which crosses the bridge of length l in time t_1 and crosses another bridge of length $2l$ in time t_2 ?

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47. [UNKNOWN] unknown

14. When 25 kg of water at 95 °C is mixed with 35 kg of water at 35 °C, the pressure being taken as constant at surrounding temperature of 15 °C, and C_p of water is 4.2 kJ/kg K, the decrease in available energy due to mixing will be nearly (a) 270.5 kJ (b) 277.6 kJ (c) 281.8 kJ (d) 288.7 kJ

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48. [UNKNOWN] unknown

15. A bar of uniform rectangular section of area A is subjected to an axial tensile load P ; its Young's modulus is E and its Poisson's ratio is μ . Its volumetric strain e_v is $\frac{P}{AE} (1 + \mu)$ $\frac{P}{AE} (1 - \mu)$ $\frac{P}{AE} (1 + 3\mu)$ $\frac{P}{AE} (1 - 3\mu)$

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49. [UNKNOWN] unknown

15. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. Average speed of the tourist during the journey is 36 km/h 33 km/h 24 km/h 18 km/h

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50. [UNKNOWN] unknown

15. A frictionless piston cylinder device contains 5 kg of steam at 400 kPa and 200 °C. The heat is now transferred to the steam until the temperature reaches 250 °C. If the piston is not attached to a shaft, its mass is constant, and by taking the values of specific volume v_1 as 0.53434 m³/kg and v_2 as 0.59520 m³/kg, the work done by the steam during this process is 121.7 kJ 137.5 kJ 153.3 kJ 189.1 kJ

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51. [UNKNOWN] unknown

16. The normal stresses on two mutually perpendicular planes are 140 N/mm² (Tensile) and 70 N/mm² (Tensile). If the maximum shear stress is 45 N/mm², the shear stress on these planes will be nearly 20.9 N/mm² 24.6 N/mm² 28.3 N/mm² 32.0 N/mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

52. [UNKNOWN] unknown

16. In a lake, the tip of a bud of lotus is seen 10 cm above the surface of water. Forced by the wind, it gradually moved, and just submerged at a distance of 30 cm. The depth of water at the root of the lotus plant will be 40 cm 50 cm 60 cm 70 cm

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53. [UNKNOWN] unknown

16. A diesel engine has a compression ratio of 14 and cutoff takes place at 6% of the stroke. The air standard efficiency

will be 74-5% 60-5% (C) 52.5% (d) 44.5%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

54. [UNKNOWN] unknown

17. The normal stresses on the two mutually perpendicular planes at a point are 120 MPa (Tensile) and 60 MPa (Tensile). If the shear stress across these planes is 30 MPa, the principal stresses will be nearly 124 MPa (Tensile) and 24 MPa (Compressive) 132 MPa (Tensile) and 24 MPa (Compressive) 124 MPa (Tensile) and 48 MPa (Tensile) 132 MPa (Tensile) and 48 MPa (Tensile)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

55. [UNKNOWN] unknown

17. A man sold a chair and a table together for Rs. 7,600, thereby making a profit of 25% on the chair and 10% on the table. By selling them together for Rs. 7,500 he would make a profit of 10% on the chair and 20% on the table. Then the cost price of chair and table will be Rs. 3000 and Rs. 4000 Rs. 3500 and Rs. 4000 Rs. 3000 and Rs. 3500 Rs. 3500 and Rs. 3500

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

56. [UNKNOWN] unknown

17. A gas mixture consists of 3 kg of O₂, 5 kg of N₂ and 12 kg of CH₄. The mass fraction and mole fraction of O₂ are 0.25 and 0.125 0.15 and 0.092 0.25 and 0.092 0.15 and 0.125

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57. [UNKNOWN] unknown

18. At a point in a material, the stresses acting on two planes at right angles to each other are : $\sigma_x = 120$ MPa and $\sigma_y = -200$ MPa and $\tau = 80$ MPa. The maximum shear stress on the element will be nearly 142 MPa 155 MPa 167 MPa 179 MPa VBY-S-CVL (4-A) ===== PAGE 5 =====

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58. [UNKNOWN] unknown

18. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be 13 cm 41 cm 52 cm 75 cm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

59. [UNKNOWN] unknown

18. An insulated pipe of 50 mm outside diameter with $\epsilon = 0.8$ is laid in a room at 30 °C. If the surface temperature is 250 °C and the convective heat transfer coefficient is 10 W/m² K, the total heat loss per unit length of the pipe will be 896-6 W/m 818-8 W/m 786-4 W/m 742-2 W/m VBY-S-MCH/50A 4 ===== PAGE 5 =====

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60. [UNKNOWN] unknown

19. The principal stresses in the wall of a container are 40 MN/mm² and 80 MN/mm². The normal makes an angle of 30° with a direction of maximum principal stress. The resultant stresses (in magnitude) in the plane will be nearly 40 MN/mm² 40 MN/mm² Higher the frequency of shaft lower will be the torque Frequency of the shaft does not influence the torque Higher the frequency of shaft higher will be the torque 84 MN/mm² 72 MN/mm² 64 MN/mm² 58 MN/mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

61. [UNKNOWN] unknown

19. Professionals who breach the 'duty of care' are liable for injuries their negligence causes. This liability is commonly referred to as

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62. [UNKNOWN] unknown

19. A wire of 8 mm diameter at a temperature of 60 °C is to be insulated by a material having $k = 174 \text{ W/m K}$. The heat transfer coefficient on the outside $h_a = 8 \text{ W/m}^2 \text{ K}$ and ambient temperature $T_a = 25 \text{ °C}$. The maximum thickness of insulation for maximum heat loss will be (a) 15.25 mm (b) 16.50 mm (c) 17.75 mm (d) 18.25 mm

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63. [UNKNOWN] unknown

20. The change in shearing force between two points on the beam is equal to the area of Loading diagram between the two points Shear force diagram between the two points Bending moment diagram between the two points M/EI diagram between the two points

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

64. [UNKNOWN] unknown

20. Information used in a business, generally unknown to the public, that the company has taken strong measures to keep confidential is called A patent A copyright A trade secret A trade mark

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

65. [UNKNOWN] unknown

20. In liquid metals, thermal boundary layer develops much faster than velocity boundary layer due to lower value of Nusselt number higher value of Prandtl number lower value of Prandtl number higher value of Nusselt number

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

66. [UNKNOWN] unknown

21. Which one of the following statements specifies shear flow? Flow of shear force along the beam It is the product of the shear stress at any level and the corresponding width b (of the section) Unbalanced force on any side of given section divided by area of section The deformation at any level due to sudden variation in shear stress 16 N/mm² 14 N/mm² 12 N/mm² 10 N/mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

67. [UNKNOWN] unknown

21. A committee of 4 is to be formed from among 4 girls and 5 boys. What is the probability that the committee will have number of boys less than number of girls ? (a) 0/30 (b) 2/4 (c) 4/4 (d) 1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

68. [UNKNOWN] unknown

21. The temperature of a body of area 0.1 m² is 900 K. The wavelength for maximum monochromatic emissive power will be nearly 2-3 μm 3.2 μm 4-1 μm 5-0 μm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

69. [UNKNOWN] unknown

22. Which one of the following statements is correct for the rotating shafts transmitting power? (a) Lower the frequency of shaft lower will be the torque

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

70. [UNKNOWN] unknown

22. The solution of initial value problem; $-3x \cdot -au = 2 - au + u$ where $u(x, 0) = 6e^{ax}$ at Professional offense Professional negligence Professional misdeed Professional malpractice (a) (1) (C) (d) $u = 6e^{-31} U = 6e^{-(2x + ii)} = 6e^{-(3x + 20)} U = 6e^{-(3x - 20)}$ 5 A - VBY-G-DST ===== PAGE 6 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

71. [UNKNOWN] unknown

22. Consider the following statements : For the laminar condensation on a vertical plate, the Nusselt theory says that

inertia force in the film is negligible compared to viscosity and weight heat flow is mainly by conduction through the liquid film, convection in liquid film as well as in vapour is neglected velocity of vapour is very high Which of the above statements are correct? 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

72. [UNKNOWN] unknown

23. The maximum shear stress induced in a solid circular shaft of diameter 15 cm, when the shaft transmits 150 kW power at 180 rpm, will be

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

73. [UNKNOWN] unknown

23. In transition boiling, heat flux decreases due to which of the following? Low value of film heat transfer coefficient at the surface during 100°C to 120°C surface temperature Major portion of heater surface is covered by vapour film which has smaller thermal conductivity as compared to liquid Nucleate boiling occurs very fast Select the correct answer using the code given below. 1 only 2 only 3 only 1, 2 and 3 VBY-S-MCH/ 50A 5 1 P.T.O. ===== PAGE 6 ===== A hemispherical furnace of radius PO m has a roof temperature of 71 = 800 K and emissivity EI = 0- 8. The flat floor of the furnace has a temperature T2 = 600K and emissivity e2 = 0-5. The view factor Fi2 from surface 1 to 2 will be (a) 0.3 (b) 0.4 (c) 0.5 (d) 0-6 Consider the following statements : Combustion chamber is the volume between TDC and BDC during the combustion process the space enclosed between the upper part of the cylinder and the top of the piston during the combustion process the space enclosed between TDC and the top of the piston during the combustion process Which of the above statements is/are correct? 1 only 2 only 3 only 1, 2 and 3 A 4-stroke diesel engine has length of 20 cm and diameter of 16 cm. The engine is producing power of 25 kW when it is running at 2500 r.p.m. The mean effective pressure of the engine will be nearly (a) 5.32 bar (b) 4.54 bar (c) 3-76 bar (d) 2-98 bar A 4-stroke, 6-cylinder gas engine with a stroke volume of 1.75 litres develops 26.25 kW at 506 r.p.m. and the MEP is 600 kN/m2. The number of misfires per minute per cylinder will be (a) 3 (b) 4 (c) 5 (d) 6

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

74. [UNKNOWN] unknown

24. A closely coiled helical spring made of 10 mm diameter steel wire has 15 coils of 100 mm mean diameter. The spring is subjected to an axial load of 100 N. For a modulus of rigidity of 816 x 104 N/mm2, the stiffness of the spring will be nearly 5-9 N/mm 6-8 N/mm 7-7 N/mm 8-8 N/mm VBY-S-CVL (5—A) . ===== PAGE 6 ===== 1000 N 1200 N 800 N 800 N

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

75. [UNKNOWN] unknown

24. If $f(z)$ has a pole of order n at $z = a$, then Residue of function $f(z)$ at a is $\text{Res } f(a) = \frac{1}{(n-1)!} \lim_{z \rightarrow a} \frac{d^{n-1}}{dz^{n-1}} \{ (z-a)^n f(z) \}$ $\text{Res } f(a) = \frac{1}{(n-1)!} \lim_{z \rightarrow a} \frac{d^{n-1}}{dz^{n-1}} \{ (z-a)^n f(z) \}$ $\text{Res } f(a) = \frac{1}{(n-1)!} \lim_{z \rightarrow a} \frac{d^{n-1}}{dz^{n-1}} \{ (z-a)^n f(z) \}$ The ratio of areas of the large circle to the small circle of a square is 4:1 1:4 2:1 1:2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

76. [UNKNOWN] unknown

25. The shear-force diagram of a beam is shown in the figure. The total of the vertically downward loads on the beam is 2600 N 3000 N 3400 N 3800N •

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

77. [UNKNOWN] unknown

25. Consider following diagram: AC is a equations is diameter of the large circle and AB=BC. au av av au (a) r w, and = dr do dr , au av av 1 au to) — = - — ana — — dr r ao ar r ao au ldvdv au (c) — = - — and — = -r ar r ao ar do au av av 1 au (d) — = r — anti = -- ar ao ar r ao

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

78. [UNKNOWN] unknown

26. A beam of triangular cross-section is subjected to a shear force of 50 kN. The base width of the section is 250 mm and the height is 200 mm. The beam is placed with its base horizontal. The shear stress at neutral axis will be nearly 2-2 N/mm² 2.7 N/mm² 3.2 N/mm² 3.7 N/mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

79. [UNKNOWN] unknown

26. Which term refers to a single person having authority to oversee all aspects of a product's production scheduling, inventory, dislocation and sales? Project management Product management Commercial management Venture management

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

80. [UNKNOWN] unknown

27. A timber beam, 100 mm wide and 150 mm deep, supports a UDL over a span of 2 m. If the safe stresses are not to exceed 28 MPa in bending and 2 MPa in shear, the maximum load that the beam can support is 16 kN/m 20 kN/m 24 kN/m 28 kN/m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

81. [UNKNOWN] unknown

27. The lowest matrix is $\begin{bmatrix} 2 & j \\ -1 & L \end{bmatrix}$ Eigen value of the 2×2 is • (a) 1 (b) 2 (c) 3 (d) 5 $f(z)$ $z = a \ar f(z)$ V113Y-G-DST - A ===== PAGE 7 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

82. [UNKNOWN] unknown

28. A 1.5 m long column has a circular cross-section of 50 mm diameter. Consider Rankine's formula with values of $1/f_c = 560 \text{ N/mm}^2$, a - for pinned ends and 1600 factor of safety of 3. If one end of the column is fixed and the other end is free, the safe load will be 9948 N 9906 N 9864 N 9822 N

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

83. [UNKNOWN] unknown

28. Consider the following statements: What is the above polymer? Mobile cranes are sophisticated machines which are designed for lifting efficiently Mobile cranes are a versatile and reliable means of lifting on site Which of the above statements is/are correct? 1 only 2 only Both 1 and 2 Neither 1 nor 2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

84. [UNKNOWN] unknown

28. Which one of the following compressors will be used in vapour compression refrigerator for plants up to 100 tonnes capacity? Reciprocating compressor Rotary compressor Centrifugal compressor Double-acting compressor

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

85. [UNKNOWN] unknown

29. A continuous beam with uniform flexural rigidity is shown in the figure. • 40 kN At B a/c 142 m 142 2 4 6 m The moment at B is 18 kNm 16 kNm 14 kNm 12 kNm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

86. [UNKNOWN] unknown

29. Which of the following statements are correct for portable step-ladders? Used on working platforms to gain height above the protected edge Used in the fully opened position Should be of a length that ensures a person's feet are not positioned any higher than the second top rung Select the correct answer using the codes given below: 2 and 3 only 1 and 3 only 1 and 2 only 1, 2 and 3 Polyamide — itride Polyacrylonitrile Polybutadiene Polyethylene

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

87. [UNKNOWN] unknown

29. A cold storage is to be maintained at -5°C while the surroundings are at 35°C . The heat leakage from the surroundings into the cold storage is estimated to be 29 kW. The actual COP of the refrigeration plant used is ird that of an ideal plant working between the same temperatures. The power required to drive the plant will be 13 kW 14 kW 15 kW 16 kW

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

88. [UNKNOWN] unknown

30. The maximum shear stress across a circular section is (1 (a) $\pm$ - Average shear stress 3 (d) (1 (c) ((b) (Average shear stress 2 -5) Average shear stress 4 Average shear stress 5 ft= 4?. VBY-S-CVL — A) ===== PAGE 7 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

89. [UNKNOWN] unknown

30. Consider the following Repeat Unit• Structure: (b) Chlorine (Cl) on oxygen (O₂) Solar Ultraviolet rays on oxygen (32) Chlorine nitrate (ClNO₃) on oxygen (O₂) 7 A — VBY-G-DST ===== PAGE 8 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

90. [UNKNOWN] unknown

30. Consider the following statements : An expansion device in a refrigeration system reduces the pressure from the condenser to the evaporator regulates the flow of the refrigerant to the evaporator depending on the load is essentially a restriction offering resistance to flow Which of the above statements are correct? 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3 VBY-S-MCH/ 50A 6 ===== PAGE 7 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

91. [UNKNOWN] unknown

31. Which of the following statements are correct in respect of temperature effect on a load-carrying three-hinged arch? No stresses are produced in a three-hinged arch due to temperature change alone. There is a decrease in horizontal thrust due to rise in temperature. There is an increase in horizontal thrust due to rise in temperature. 1 and 2 only 1 and 3 only 2 only 3 only D14-4 m-1.14— 4m-41 The magnitude of the horizontal support reaction at E is 400 kN 300 kN 250 kN 200 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

92. [UNKNOWN] unknown

31. Which of the following measures is/are correct for using Mobile Equipment Working Platform (MEWP) ? Tyres are properly inflated and air filled SWL to be marked in platforms as identification for carrying loads 1 only 2 only Both 1 and 2 Neither 1 nor 2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

93. [UNKNOWN] unknown

31. A reversed Carnot engine is used for heating a building. It supplies 210×10^3 kJ/hr of heat to the building at 20°C . The outside air is at -5°C . The heat taken from the outside will be nearly 192×10^3 kJ/hr 188×10^3 kJ/hr 184×10^3 kJ/hr 180×10^3 kJ/hr

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

94. [UNKNOWN] unknown

32. Consider the frame as shown in the figure. HE 200N 2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

95. [UNKNOWN] unknown

32. Ozone layer present in the atmosphere protects life on earth by not permitting harmful radiations present in the fl sunlight to penetrate through it. Ozone layer is formed by the reaction of (a) Chlorofluorocarbons (CFCs) on oxygen (O₂)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

96. [UNKNOWN] unknown

32. In an Electrolux refrigerator, a thermo- siphon bubble pump is used to lift the weak aqua solution from the generator to the separator weak aqua solution from the separator to the absorber strong aqua solution from the generator to the separator strong aqua solution from the generator to the evaporator

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

97. [UNKNOWN] unknown

33. The load system in the figure moves from left to right on a girder of span 10 m. 150 kN 120 kN 60 kN A4 m The maximum bending moment for the girder is nearly 820 kNm 847 kNm 874 kNm 890 kNm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

98. [UNKNOWN] unknown

33. The enthalpy of moist air with normal notations is given by $h = (1.005 + 1.88W) t + 2500W$ $h = 1.88Wt + 2500W$ $h = 1.005Wt$ $h = (188 + 1005W) t + 2500W$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

99. [UNKNOWN] unknown

34. Two wheel loads 80 kN and 200 kN respectively spaced 2 m apart, move - on a girder of span 16 m. Any wheel load can lead the other. The maximum negative shear force at a section 4 m from the left end will be —50 kN —60 kN —70 kN —80 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

100. [UNKNOWN] unknown

34. A cone resting on its base in horizontal plane (HP) is cut by a plane inclined to the axis and parallel to one of its generators, the sectional view will be Ellipse Parabola Hyperbola Circle Which of the above statements is/are correct? 1 only 2 only Both 1 and 2 Neither 1 nor 2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

101. [UNKNOWN] unknown

34. If the relative humidity of atmospheric air is 100%, then the wet-bulb temperature will be more than dry-bulb temperature equal to dew-point temperature equal to dry-bulb temperature less than dry-bulb temperature During an air-conditioning of a plant, the room sensible heat load is 40 kW and room latent heat load is 10 kW, ventilation air is 25% of supply air. At full load, the room sensible heat factor will be (a) 0.9 (b) 0.8 (c) 0.7 (d) 0.6 A 2-stroke oil engine has bore of 20 cm, stroke 30 cm, speed 350 r.p.m., i.m.e.p. 275 IEN/m², net brake load 610 N, diameter of brake drum 1 m, oil consumption 4.25 kg/hr, calorific value of fuel 44 x 10³ kJ /kg. The indicated thermal efficiency will be (a) 29.1% (b) 3F3% (c) 33-5% (d) 35.7% The hydraulic efficiency of a turbine is the ratio of (a) mechanical energy in the output shaft at coupling and hydrodynamic energy available from the fluid (13) mechanical energy supplied by the rotor and hydrodynamic energy available from the fluid useful hydrodynamic energy in the fluid at final discharge and mechanical energy supplied to the rotor useful hydrodynamic energy in the fluid at final discharge and mechanical energy supplied to the shaft and coupling VBY-S-MCH/50A 7 [P.T.O. ===== PAGE 8 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

102. [UNKNOWN] unknown

35. The maximum possible span for a cable supported at the ends at the same level (assuming it to be in a parabolic profile) allowing a central dip of —1 of the span with 10 permissible stress of 150 N/mm² (where the steel weighs 78,000 N/m³) will be nearly 1270 m 1330 m 1388 m 1450 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

103. [UNKNOWN] unknown

35. Consider the following components: Knowledge of psychology Knowledge of the theory of varia- tion Knowledge of process Knowledge of the system and the theory of optimization

104. [UNKNOWN] unknown

36. A three-hinged arch has a span of 30 m and a rise of 10 m. The arch carries UDL of 60 kN/m on the left half of its span. It also carries two concentrated loads of 160 kN and 100 kN at 5 m and 10 m from the right end. The horizontal thrust will be nearly 446 kN 436 kN 428 kN 418 kN

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105. [UNKNOWN] unknown

36. Consider the following statements: CAD' to insert Objects in the current file Objects in any file Blocks in any drawing file Blocks and wblocks in the current drawing Greenfield Privatization or Incremental Privatization denotes encouragement to private sector in areas hitherto reserved for Public Enterprises Cold Privatization refers to measures taken to distance Public Enterprises from the Government

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

106. [UNKNOWN] unknown

37. An unstable vibratory motion due to combined bending and torsion which occurs in flexible plate like structures is called Galloping Owalling Mutter Oscillation 70 kN VBY-S-CVL (7—A) ===== PAGE 8 ===== A propped cantilever beam of span / and constant plastic moment capacity M. carries a concentrated load at mid-span. The load at collapse will be 2 MP 4 M 6 MP 1 (d) A steel plate is subjected to tension. The tensile force is applied over a width 'a' whereas the gross width of the plate is 'b'. The dispersion of the force from the point of application is at about 30° with the axis and extends to a maximum width of 12 times the thickness t of the plate. The effective width which comes into action will be 2a + 12t a + 12t a + 24t 2a + 24t A wind brace is to be provided between two columns spaced at 5 m, at an inclination of 30° with the horizontal, to resist a tension of 320 kN developed by a wind force. The effective area required will be nearly (considering 150 N/m² as a relevant factor) 1670 mm² 1640 mm² 1600 mm² 1570 mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

107. [UNKNOWN] unknown

37. Which of the following steps are involved in the product improvement cycle? Sell it in the market Determine quality of performance Design the product based on customer needs Test it in the laboratory Which of the above components comprise the basis of Deming's Systems of Profound Knowledge? Select the correct answer using the codes given below: (a) 1, 2 and 3 only (a) 1 and 3 only (b) 1, 3 and 4 only (b) 2 and 4 only (c) 1, 2 and 4 only (c) 1, 3 and 4 only (d) 2, 3 and 4 only (d) 1, 2, 3 and 4 VBY-G-DST — A 8 ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

108. [UNKNOWN] unknown

38. Who is responsible for establishing, documenting and maintaining procedures for post-production handling functions such as storage, packaging and delivery? Production Manager Marketing Manager Vendor Quality Supervisor

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

109. [UNKNOWN] unknown

38. Consider the following statements regarding compounding in steam turbines :

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

110. [UNKNOWN] unknown

39. A unit produces packing boxes. Out of hourly production of 4,000 boxes, 20 were found to be non-conforming. If the inspector randomly chooses a box from an hour's production, the probability of it being non-conforming is 0-02 0-10 0-005 0-05

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

111. [UNKNOWN] unknown

39. In a lawn sprinkler, water leaves the jet with an absolute velocity of 2 m/s and the sprinkler arms are 0.1 m in length.

The sprinkler rotates at a speed of 120 r.p.m. The utilization factor of this sprinkler will be nearly 0.72 0.64 0.56 0.49 In axial flow pumps and compressors, the combined velocity diagram with common base is used to determine change in absolute velocity ($V_2 - V_1$) relative velocity ($U_2 - U_1$) whirl velocity ($V_{u2} - V_{u1}$) In a steam turbine with steam flow rate of 1 kg/s, inlet velocity of steam of 100 m/s, exit velocity of steam of 150 m/s, enthalpy at inlet of 2900 kJ/kg, enthalpy at outlet of 1600 kJ/kg, the power available from the turbine will be nearly 1575.5 kW 1481.6 kW 1387.7 kW 1293.8 kW VBY-S-MCH/ 50A 8 ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

112. [UNKNOWN] unknown

40. Which of the following are relevant factors regarding quality in service sector? Timeliness of service Customer participation Company personnel motivation Company culture Select the correct answer using the codes given below: 1, 3 and 4 only 1, 2 and 3 only 1, 2, 3 and 4 2, 3 and 4 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

113. [UNKNOWN] unknown

40. Which one of the following statements is correct with respect to axial flow 50% reaction turbine? In impulse turbine, steam pressure remains constant between ends of the moving blades. In reaction turbine, steam pressure drops from inlet to outlet of the blade. In velocity compounding, partial expansion of steam takes place in the nozzle and further expansion takes place in the rotor blades. The combined velocity diagram is symmetrical. The outlet absolute velocity should not be axial for maximum utilization. Angles of both stator and rotor are not identical. (d) For maximum utilization, the speed U : U_2 ratio $\sin \alpha$ = U_1 Which of the above statements are correct? 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

114. [UNKNOWN] unknown

41. A beam column for a non-sway column in a building frame is subjected to a factored axial load of 500 kN, factored moment at bottom of column of 45 kNm. For ISHB 200, the values are $A = 4750 \text{ mm}^2$, $y_y = 451$, $h = 200 \text{ mm}$, $b = 200 \text{ mm}$, $b_f = 9 \text{ mm}$ and the effective length is $0.8 L$. Its buckling load will be 910 kN 930 kN 950 kN 980 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

115. [UNKNOWN] unknown

41. Which one of the following tests can be resorted to in order to check the structural soundness conformance to the specified standards, when all other tests fail? Destructive Non-destructive Full scale load Masonry

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

116. [UNKNOWN] unknown

42. Which of the following assumptions are correct for ideal beam behaviour? The compression flange of the beam is restrained from moving laterally. The tension flange of the beam is restrained from moving laterally. Any form of local buckling is prevented. 2 and 3 only 1 and 3 only 1 only 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

117. [UNKNOWN] unknown

42. Which of the following are the sources of variation in quality control process in construction? Material Operator Inspection activity Select the correct answer using the codes given below: 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

118. [UNKNOWN] unknown

43. In which one of the following industrial roofing contexts, is the loading carried by the combination of pure flexure and flexure due to shear induced by the relative deformation between the ends of the top and bottom chord members? Vierendeel girders Scissors girders Lenticular girders Mansard girders

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

119. [UNKNOWN] unknown

43. What is the break-even sale for the following products? Products A B C Sales (Units) 5,000 6,000 4,000 Unit selling price (Rs.) 10 15 18 Unit variable price (Rs.) 6 4 13 Fixed cost of the product is (Rs. 20 000) Rs. 90,000 Rs. 80,000 Rs. 60,000 Rs. 40,000 9 A — VBY-G-DST ===== PAGE 10 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

120. [UNKNOWN] unknown

43. In an isentropic flow through a nozzle, air flows at the rate of 600 kg/hr. At inlet to nozzle, the pressure is 2 MPa and the temperature is 127 °C. The exit pressure is of 0.5 MPa. If the initial velocity of air is 300 m/s, the exit velocity will be (a) 867 m/s (b) 776 m/s (c) 685 m/s (d) 594 m/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

121. [UNKNOWN] unknown

44. Bearing stiffeners are provided At the ends of plate girders At the ends of plate girder and on both faces of the web At the ends of plate girder and only on one face of the web At the points of concentrated loads, to protect the web from the direct compressive loads 8 MP 0 VBY-S-CVL (8 —A) ===== PAGE 9 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

122. [UNKNOWN] unknown

44. Which of the following approaches are correct regarding total quality? Opportunity to improve Adoption requires little change React to competitive threats Select the correct answer using the codes given below: (a) 1 and 2 only (b) 1 and 3 only (c) 2 and 3 only A transient labour force The construction process is relatively short in duration Hierarchical and vertical organization structure The construction process has not focused on the detailed needs of the customer Select the correct answer using the codes given below: 1 and 4 only 2 and 3 only 1 and 2 only 3 and 4 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

123. [UNKNOWN] unknown

44. In a steam turbine, the nozzle angle at the inlet is 18°. The relative velocity is reduced to the extent of 6% when steam flows over the moving blades. The output of the turbine is 120 10/kg flow of steam. If the blades are equiangular, the speed ratio and the absolute velocity of steam at inlet for maximum utilization are nearly 1142 and 230.2 m/s 048 and 230.2 m/s 042 and 5151 m/s 048 and 5151 m/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

124. [UNKNOWN] unknown

45. If the cost of purlins/unit area is p and the cost of roof covering/unit area is r, then cost of trusses/unit area / for an economical spacing of the roof trusses will be p + r 2p + r p + 2r 2p + 2r

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

125. [UNKNOWN] unknown

45. Which of the following are constraints to the use of TQM in construction (c) process? VBV-G-DST — A 10 ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

126. [UNKNOWN] unknown

45. An air compressor compresses atmospheric air at 01 MPa and 27 °C by 10 times of air inlet pressure. During compression, the heat lost to the surrounding is estimated to be 5% of compression work. Air enters the compressor with a velocity of 40 m/s and leaves with 100 m/s. The inlet and exit cross-sectional areas are 100 cm² and 20 cm² respectively. The temperature of air at the exit from the compressor will be 1498 K 1574 K 1654 K 1726 K

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

127. [UNKNOWN] unknown

46. A welded plate girder of span 25 m is laterally restrained throughout its length. It has to carry a load of 80 kN/m over the whole span besides its weight. If $K = 200$ and $f = 250$ MPa, the thickness of web will be nearly 10 mm 14 mm 16 mm 20 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

128. [UNKNOWN] unknown

46. BOD of a waste water sample is estimated to be 180 mg/l. Assuming 4 mg/l BOD can be consumed in the BOD bottle, the volume of undiluted sample to be added to a 300 ml bottle is nearly 6-7 ml 5.6 ml 4-4 ml 3.3 ml

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

129. [UNKNOWN] unknown

46. A compressor delivers 4 m³ of air having a mass of 5 kg. The specific weight and specific volume of air being delivered will be nearly (a) 12.3 N/m³ and 0.8 m³/kg (b) 14.6 N/m³ and 0.4 m³/kg (c) 12.3 N/m³ and 0.4 m³/kg (d) 14.6 N/m³ and 0.8 m³/kg

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

130. [UNKNOWN] unknown

47. A propped cantilever ABCD is loaded as shown in figure. If it is of uniform cross-section, the collapse load of the beam will be nearly 8 kN 16 kN 24 kN 32 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

131. [UNKNOWN] unknown

47. Venturi scrubber, a device used to remove particulate matter from the atmosphere, works on the principle of Settling by gravitational force Removal by centrifugal force Removal by electrically charged particles (d) Removal by atomized water vapour

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

132. [UNKNOWN] unknown

47. In centrifugal compressors, there exists a loss of energy due to the mismatch of direction of relative velocity of fluid at inlet with inlet blade angle. This loss is known as frictional loss incidence loss clearance loss leakage loss

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

133. [UNKNOWN] unknown

48. Consider a triangular section with base b and height h as shown in the figure. The shape factor will be nearly 2.3 3.2 4.1 5.0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

134. [UNKNOWN] unknown

48. Environmental Impact Assessment (EIA) is aimed to help Estimate future needs of the society Smooth implementation of a project Cope with rapid increase in population Resource conservation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

135. [UNKNOWN] unknown

48. A centrifugal compressor develops a pressure ratio of 5 and air consumption of 30 kg/s. The inlet temperature and pressure are 15 °C and 1 bar respectively. For an isentropic efficiency of 0.85, the power required by the compressor will be nearly 5964 kW 5778 kW 5586 kW 5397 kW

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

136. [UNKNOWN] unknown

49. Fatigue in RCC beams will not be a problem if the number of cycles is less than 20,000 25,000 (c). 30,000 (d) 35,000

137. [UNKNOWN] unknown

49. Which one of the following is a terrestrial type of ecosystem? Limnetic Estuary Prairie Reefs (d) 1, 2 and 3 (a) (b)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

138. [UNKNOWN] unknown

49. The efficiency of superheat Rankine cycle is higher than that of simple Rankine cycle because the enthalpy of main steam is higher for superheat cycle the mean temperature of heat addition is higher for superheat cycle the temperature of steam in the condenser is high the quality of steam in the condenser is low

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

139. [UNKNOWN] unknown

50. The desired characteristic strength of a mix is 20 N/mm². The standard deviation is 4 N/mm² for 150 mm size of concrete cubes; and K = 1.645. The average strength of the cubes will be nearly 38.2 N/mm² 32.4 N/mm² 26.6 N/mm² 22.8 N/mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

140. [UNKNOWN] unknown

50. What are the limitations of solar energy? Collecting solar energy over large areas and converting it to other forms that can be conveniently transported, stored and used in existing equipment is not economical Low density of solar energy as compared to coal, oil and gas Its major applications are photo-thermal conversion, solar water heating, green housing technology and photo voltaic conversion Select the correct answer using the codes given below: 1, 2 and 3 1 and 2 only. 1 and 3 only 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

141. [UNKNOWN] unknown

50. In steam power cycle, the process of

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

142. [UNKNOWN] unknown

51. A circular column is subjected to an un-factored load of 1600 kN. The effective length of the column is 3.5 m, the concrete is M25, and the value of $p_g = \frac{A_s}{A_g} = 2\%$ for Ag Fe 415 steel. The design diameter of the column will be nearly 446 mm 432 mm (c) 424 mm (d) 410 mm A VBY-S-CVL (9—A) ===== PAGE 10 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

143. [UNKNOWN] unknown

51. Acid rain results when gaseous emissions of Sulfur oxides (SO) and nitrogen oxides (NO) interact with water vapour and Moonlight, and are chemically converted to strong acidic compounds such as sulfuric acid (H₂SO₄) and nitric acid (HNO₃) Sunlight, and are chemically converted to strong acidic compounds such as sulfuric acid (H₂SO₄) and nitric acid (HNO₃) Moonlight, and are chemically converted to weak acidic compounds such as sulfuric acid (H₂SO₄) and nitric acid (HNO₃) Sunlight, and are chemically converted to weak acidic compounds such as sulfuric acid (H₂SO₄) and nitric acid (HNO₃)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

144. [UNKNOWN] unknown

51. The internal irreversibility of Rankine cycle is caused by (d) 16.9 kg/kg of fuel fluid friction throttling

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

145. [UNKNOWN] unknown

52. A strut is made of a circular bar, 5 m long and pin-jointed at both ends. When freely supported the bar gives a mid-span deflection of 10 mm under a load of 80 N at the centre. The critical load will be 8485 N 8340 N 8225 N 8110 N

146. [UNKNOWN] unknown

52. The Vilinamata Tragedy' was caused by the eating of fish growing in the Minamata Bay contaminated with Peroxy alylnitrate Methyl isocyanate Potassium cyanide Methylmercury

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

147. [UNKNOWN] unknown

52. A 1 g sample of fuel is burned in a bomb calorimeter containing 1.2 kg of water at an initial temperature of 25 °C. After the reaction, the final temperature of the water is 33.2 °C. The heat capacity of the calorimeter is 837 J/ °C. The specific heat of water is 4.18 J/ °C. The heat released by the fuel will be nearly 36 kJ/g 42 kJ/g 48 kJ/g 54 kJ/g

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

148. [UNKNOWN] unknown

53. The recommended imposed load on staircase in residential buildings as per IS 875 is 5.0 kN/m² 3-0 kN/m² . F5 kN/m² F3 kN/step

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

149. [UNKNOWN] unknown

53. What are the advantages of Biomass as a source of energy? Its storage and transportation is possible It is ecologically safe and is inoffensive Can be developed with present man and material abilities Low capital input required Select the correct answer using the codes given below: 1, 2, 3 and 4 1, 2 and 3 only 3 and 4 only 3 and 4 only 11 A — VBY-G-DST ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

150. [UNKNOWN] unknown

53. A boiler is having a chimney of 35 m removing non-condensable gases is height. The draught produced in terms called of water column is 20 mm. The temperature of flue gas inside the (a) scavenging process chimney is 365 °C and that of air (b) deaeration process outside the chimney is 32 °C. The mass of air used will be nearly (c) exhaust process (a) 10.3 kg/kg of fuel (d) condensation process (b) 12.5 kg/kg of fuel 14/ kg/kg of fuel

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

151. [UNKNOWN] unknown

54. A 230 mm brick masonry wall is •to be provided with a reinforced concrete footing on site having soil with safe bearing capacity of 125 kN/m², unit weight of 17.5 kN/m³ and angle of shearing resistance of 30°. The depth of footing will be nearly 0.8 m 0.7 in 0.6 m 0.5 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

152. [UNKNOWN] unknown

54. Consider the following data for a domestic biogas plant: Number of cows = 5 Retention time = 20 days Temperature = 30°C Dry matter consumed = 2 kg/day Biogas yield = 0-24 m³/kg Efficiency of burner = 60% Methane proportion = 0-8 Heat of combustion of Methane = 28 MJ/m³ Density of dry material in fluid = 50 kg/m³ The power available from the digester will be nearly 16-2 MJ/day 243 MJ/day 323 MJ/day 48-6 MJ/day

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

153. [UNKNOWN] unknown

54. A 2 kg of steam occupying 0.3 m³ at mixing 15 bar is expanded according to the law Select the correct answer using the code given below, pV^{1.3} = constant to a pressure of 1.5 bar. The work done during the expansion will be (a) 1 and 2 only (a) 602.9 kJ (b) 1 and 3 only 6061 kJ 2 and 3 only (c) 612.5 kJ (d) 1, 2 and 3 (d) 618.3 kJ VBY-S-MCH/50A 10 ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

154. [UNKNOWN] unknown

55. A rectangular beam 200 mm wide has an effective depth of 350 mm. It is subjected to a bending moment of 24,000 Nm. The permissible stresses are $c = 5 \text{ N/mm}^2$, $t = 140 \text{ N/inm}^2$; and m is 18. The required area of tensile reinforcement will be 688 mm² 778 mm² 864 mm² 954 mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

155. [UNKNOWN] unknown

55. The best tool to ensure that there is neither piling up of stocks nor shortage of materials in a project to run it economically is Economic Order Quantity ABC Analysis Inventory Control and Management Gantt Chart Method

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

156. [UNKNOWN] unknown

55. Which of the following statements is/are correct regarding superheater in boilers? It is a heat exchanger in which heat is transformed to the saturated steam to increase its temperature. It raises the overall efficiency. It reduces turbine internal efficiency. Select the correct answer using the code given below. 1 and 2 1 and 3 2 and 3 1 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

157. [UNKNOWN] unknown

56. Which of the following statements are correct with reference to ensuring minimum shrinkage of prestressed concrete? The water-cement ratio and proportion of cement paste should be kept minimum to reduce shrinkage. Aggregates of larger size, well graded for minimum void, need a smaller amount of cement paste, and attendant shrinkage will be smaller. Harder and denser aggregates of low water absorptions and high modulus of elasticity will exhibit small shrinkage. 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

158. [UNKNOWN] unknown

56. A machine is expected to generate cash saving (after-tax) of Rs. 50,000 per annum over a period of 5 years. Salvage value of machine is 40% of the original cost. If accounting rate of return is 20%, cost of two such machines will be Rs. 78,125 Rs. 1,56,250 Rs. 3,12,500 Rs. 6,25,000

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

159. [UNKNOWN] unknown

56. Water vapour at 90 kPa and 150 °C enters a subsonic diffuser with a velocity of 150 m/s and leaves the diffuser at 190 kPa with a velocity of 55 m/s, and during the process, 1.5 kJ/kg of heat is lost to the surrounding. For water vapour, C_p is 21 kJ/kg K. The final temperature of water vapour will be (a) 154°C (b) 158 °C (c) 162 °C (d) 166°C

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

160. [UNKNOWN] unknown

57. During earthquakes, the corner and edge columns may be subjected to Uniaxial bending Biaxial bending Combined biaxial bending and torsion Combined biaxial bending and tension

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

161. [UNKNOWN] unknown

57. It is expected to receive Rs. 5,000 annually for 3 years with each receipt occurring at the end of the year. With a discount rate of 10%, the present value of the annuity will be nearly (a) Rs. 12,435 (13) Rs. 9,945 Rs. 4,975 Rs. 2,487

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

162. [UNKNOWN] unknown

57. A steam turbine is supplied with steam at a pressure of 20 bar gauge. After expansion in the steam turbine, the steam passes to condenser which is maintained at a vacuum of 250 mm of mercury by means of pumps. The inlet and exhaust steam pressures will be nearly 2101 kPa and 68 kPa 2430 kPa and 78 kPa 2101 kPa and 78 kPa 2430 kPa and 68 kPa

163. [UNKNOWN] unknown

58. The minimum number of bars required in a rectangular column for an earthquake resistant design, is 4 6 8 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

164. [UNKNOWN] unknown

58. In a project life cycle, the maximum percentage of effort is done in Concept phase Definition phase Planning and organizing phase Implementation phase

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

165. [UNKNOWN] unknown

58. In a power plant, the efficiencies of the electric generator, turbine, boiler, thermodynamic cycle and the overall plant are 0.97, 0.95, 0.92, 0.42 and 0.33 respectively. The total electricity generated for running the auxiliaries will be nearly (a) +9% (c) 6-5%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

166. [UNKNOWN] unknown

59. The permissible or allowable compressive stress f_a , of brick masonry does not depend on Type and strength of bricks Efflorescence of bricks Strength of mortar Slenderness ratio

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

167. [UNKNOWN] unknown

59. In progress of a project, the percentage of error will be less in Definitive cost estimate Detailed estimate Preliminary estimate Study estimate V13Y-G-DST — A 12 ===== PAGE 13 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

168. [UNKNOWN] unknown

59. A turbine in which steam expands both in nozzle as well as in blades is called as impulse reaction turbine reciprocating steam turbine gas turbine Curtis turbine

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

169. [UNKNOWN] unknown

60. A masonry dam 8 m high, 1.5 m wide at the top and 5 m wide at the base retains water to a depth of 7.5 m, the water face of the dam being vertical. If the weight of water is 9.81 kN/m³, weight of masonry is 22 kN/m³, the maximum intensity of stress developed at the base will be nearly 196 kN/m² 182 kN/m² 160 kN/m² 148 kN/m² VBY-S-CVL (10 — A) ===== PAGE 11 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

170. [UNKNOWN] unknown

60. In principle, the network should not be made complex. No control system, for that matter, can operate unless it is kept simple. This principle is called CPM PERT KISS GERT

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

171. [UNKNOWN] unknown

60. Consider the following statements regarding reaction turbine : Blade shape is aerofoil type and its manufacturing is difficult. It is suitable for small power. Leakage losses are less compared to friction losses. Which of the above statements is/are correct? 1 only 2 only 3 only 1, 2 and 3 VBY-45-4VICH/50A 11 [P.T.O. ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

172. [UNKNOWN] unknown

61. A front-end loader on a given job moves a load of 1.5 m³ of loose soil in one cycle consisting of

loading-lifting-travelling-unloading-return trip-and-ready for next loading. If each cycle time is 1.2 minutes, the actual output will be 75 m³/hour 70 m³/hour 65 m³/hour 60 m³/hour

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

173. [UNKNOWN] unknown

61. Which one of the following is a viable alternative to term-loans and are instruments for raising debt finance by large publicly traded firms? Shares Debentures Asset loans Gold loans

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

174. [UNKNOWN] unknown

61. The solar heat pipe works on the principle of heating and condensation cycle evaporation and condensation cycle cooling and condensation cycle heating and evaporation cycle

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

175. [UNKNOWN] unknown

62. Which of the following techniques belong to 'Project Time Plan'? Critical path method Precedence network analysis Line of balance technique Linear programme chart 1, 2 and 3 only 1,2 and 4 only 3 and 4 only 1, 2, 3 and 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

176. [UNKNOWN] unknown

62. Which one of the following makes the design, assembly and operation of complex systems feasible and practical? System Architecture Modularization Standardization Composition

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

177. [UNKNOWN] unknown

62. A good approximation of the measured solar spectrum is made by black-body energy distribution Planck's energy distribution inverse square law solar constant

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

178. [UNKNOWN] unknown

63. A construction equipment has an initial cost of 2,00,000 and salvage value of? 50,000 at the end of an economic life of 5 years. The rate of straight-line depreciation and total depreciation will be 0.1 and? 1,50,000 0-2 and? 1,50,000 0.1 and? 1,00,000 0.2 and 1,00,000 VBY-S-CVL (11 — A.) Compaction Consolidation Grading Ditch digging

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

179. [UNKNOWN] unknown

63. Which one of the following schedules shows the specific activities necessary to complete an activity or work package? Project schedule Master schedule Task schedule Internal schedule

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

180. [UNKNOWN] unknown

63. Which one of the following types of tracker uses liquid contained in canisters that can turn easily into vapour? Active tracker Passive tracker Single-axis tracker Altitude-azimuth tracker

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

181. [UNKNOWN] unknown

64. Consider the following assembly with different operations : Operation Standard time, minutes A 60 B 65 C 29 D 37 E 28 F 63 G 36 H 126 K 64 There are 250 working days in a year to produce 4000 units in a year. The minimum number of work stations required will be 13 12 11 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

182. [UNKNOWN] unknown

64. In a stable ceramic crystal structure, a cation is surrounded by three anions in the form of a planar equilateral triangle. The ratio of the cation-anion radius for the crystal is nearly 0.16 0.24 0.32 0.41

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

183. [UNKNOWN] unknown

64. Which type of flat-plate collector is used to heat the swimming pools with plastic panel, utilizing solar energy? Pipe and fin type Full water sandwich type Thermal traps type Corrugated plate with selective surface type

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

184. [UNKNOWN] unknown

65. Flattening and smoothing the road surface by scrapping is called

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

185. [UNKNOWN] unknown

65. During tensile testing of a material, if cross-sectional area of the specimen is doubled, the load required to produce the same elongation shall be Double Half Same Four times

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

186. [UNKNOWN] unknown

65. The edge loss U_e in a solar collector with respect to edge area A_s , collector area A , and back loss coefficient U_b is

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

187. [UNKNOWN] unknown

66. The amount of time by which the start of the activity may be delayed without interfering with the start of any succeeding activity is called Activity float Free float Total float Interfering float ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

188. [UNKNOWN] unknown

66. When two or more chemically different monomers are polymerized to form a cross link polymer along with some byproduct such as water, the process is known as Crystallographic polymerization Additive polymerization Copolymerization Condensation polymerization 13 A — VBY-G-DST ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

189. [UNKNOWN] unknown

66. In solar porous type air heater, the pressure drop is usually higher than non-porous type same as in non-porous type lower than non-porous type zero

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

190. [UNKNOWN] unknown

67. A crew consisting of two carpenters and one helper can fix 10 m² of a slab form work in hours and the hourly labour rate of a carpenter is 85 and for a helper is 69-50. An average hourly rate per worker of the crew will be nearly 90 80 70 60

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

191. [UNKNOWN] unknown

67. The number of atoms per unit length whose centres lie on the direction vector for a specific crystallographic direction is called Linear density Theoretical density Atomic density Avogadro number

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

192. [UNKNOWN] unknown

67. In a drainback solar water heating system the water in the heat exchanger is recycled the water is heated in

collectors only during times when there is available heat at the collector, the mixture of water and propylene-glycol is heated and returned to a solar storage tank there is an expansion tank with enclosed air chamber to assist water draining

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

193. [UNKNOWN] unknown

68. A project with the production cost of 100 crores, has 20,000 man-months as direct labour, of which 60% is non-productive time The labour cost as estimated while tendering is 20% of project cost. If 15% of the wastage resulting from non-productive time is eliminated by using improved methods, the resulting saving in labour cost will be 14.5% 18-5% 22-5% 26-5%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

194. [UNKNOWN] unknown

68. Which of the following features of atoms determine the degree to which the solute atoms dissolve in the solvent atoms? Atomic size factor Crystal structure Electronegativity 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

195. [UNKNOWN] unknown

68. A PV cell is illuminated with irradiance (E) of 1000 W/m². If the cell is 100 mm x 100 mm in size and produces 3 A at 0.5 V at the maximum power point, the conversion efficiency will be (a) 15% (b) 19% (c) 23% (d) 27% VSY-S-MCH/50A 12 ===== PAGE 13 ===== 80 mm 10 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

196. [UNKNOWN] unknown

69. Consider the following data: Work is carried out by a contractor employing labour with 25% overtime per day Working for 5 days a week Contractor peak manpower is 40 per day Build-up period is 20% Rundown period is 10% Total effort in standard man days is 1200 The duration of work by Trapezoidal manpower distribution pattern will be 5-5 weeks 6.5 weeks 7-5 weeks 8-5 weeks

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

197. [UNKNOWN] unknown

69. A state for ionic compounds wherein there is the exact ratio of cations to anions as predicted by the chemical formula is Electroneutrality Stoichiometry Equilibrium defect Frankel defect

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

198. [UNKNOWN] unknown

69. In a barrage of 300000 m² area with a tide height of 3 m, barrage drain time of 10 hr, density of seawater as 1025 kg/m³ and gravitational acceleration as 9.8 m/s², the average power will be

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

199. [UNKNOWN] unknown

70. A systematic measurement and evaluation of the way in which an organization manages its health and safety programme against a series of specific and attainable standards is called Safety inspection Safety audit Safety plan Safety committee

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

200. [UNKNOWN] unknown

70. The capacity of a material to absorb energy when it is deformed elastically and then, upon unloading, to have this energy recovered is called Ductility Tensile strength Elasticity Resilience

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

201. [UNKNOWN] unknown

70. The platinum nano-coating is made on the anode of the fuel cell to (a) create lighter and more efficient fuel cell membranes

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

202. [UNKNOWN] unknown

71. On a construction project, the contractor, on an average, employed 100 workers with 50 hours working per week. The project lasted for 35 weeks and, during this period, 14 injuries occurred. The disabling injury-frequency rate will be (based on one lakh of man hours worked) (a) 5 (b) 6 (c) 7 (d) 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

203. [UNKNOWN] unknown

71. In which one of the following phase transformations, there are no compositional alterations? Incongruent transformations Congruent transformations Non-equilibrium transformations Equilibrium transformations

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

204. [UNKNOWN] unknown

71. In a fuel cell, electric current is produced when (a) hydrogen and oxygen react with each other and electrons are freed (b) hydrogen reacts with water and The in the (a) μ_{H_2} $\rightarrow$ $\mu_{\text{H}_2\text{O}}$ maximum bending stress produced bar at xx is nearly 20.5 N/mm² electrons are freed (c) oxygen reacts with water and 18.8 N/mm² electrons are freed 16.3 N/mm² electrons react with molecules of hydrogen and oxygen is freed (d) 14.5 N/mm² VBY-S-MCH/50A 13 [P.T.O. ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

205. [UNKNOWN] unknown

72. The graphical representations wherein long duration jobs are broken down to key segmental elements, wherein events are shown in chronological order without attention to logical sequencing, and wherein interdependencies between the events is not highlighted, is referred to as CPM Milestone chart GANTT chart PERT VBY-S-CVL (2 — A ===== PAGE 13 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

206. [UNKNOWN] unknown

72. In a simple cubic structure, atomic power factor is nearly 0-9 0-7 (c) 0-5 (d) 0-3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

207. [UNKNOWN] unknown

72. Which one of the following is suitable for fuel cell electric vehicle (FCEV)? Direct methanol fuel cell (DMFC) Alkaline fuel cell (AFC) 377 kW (c) Proton exchange membrane fuel 381 kW cell (PEMFC) 388 kW (d) Solid oxide fuel cell (SOFC) 396 kW

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

208. [UNKNOWN] unknown

73. A ship weighs 127 MN. On filling the ship's boats on one side with water weighing 600 kN with the mean distance of the boats from the centre line of the ship being 10 m, the angle of displacement of the plumb line is 2°16'. The metacentric height will be nearly (Take $\sin 2^\circ 16' = 0.04$, $\cos 2^\circ 16' = 0.9992$ and $\tan 2^\circ 16' = 0.04$) F73 m F42 m 1.1.8 m 0.87 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

209. [UNKNOWN] unknown

73: Which of the following are the advantages of coding audiovisual objects? It allows interaction with the content It improves reusability and coding the content It allows content-based scalability 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3 VBY-G-DST — A 14 ===== PAGE 15 =====

210. [UNKNOWN] unknown

73. A pull of 100 lN acts on a bar as shown in the figure in such a way that it is parallel to the bar axis and is 10 mm away from n : make the fuel effective create high thermal conductivity in the cell make the fuel cell non-corrosive

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

211. [UNKNOWN] unknown

74. For frictionless adiabatic flow of compressive fluid, the Bernoulli's equation with usual notations is $k P_1 V_1^{1/k} - \frac{1}{k-1} \frac{V_1^2}{g} + \frac{P_1}{\rho g} + z_1 = k P_2 V_2^{1/k} - \frac{1}{k-1} \frac{V_2^2}{g} + \frac{P_2}{\rho g} + z_2$ (C)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

212. [UNKNOWN] unknown

74. The transmission of real-time streams across networks uses Bandwidth Allocation Mechanism (BAM), which is based on Stream peak rate Bucket rate Token bucket depth Packet size

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

213. [UNKNOWN] unknown

74. The frequency of oscillation is the number of cycles per unit time described by the particle, given by the relation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

214. [UNKNOWN] unknown

75. The phenomenon of generation of lift by rotating an object placed in a free stream is known as Coanda effect Magnus effect Scale effect Buoyancy effect

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

215. [UNKNOWN] unknown

75. The quality of service provided in a computer network is Presentation layer issue Session layer issue Network layer issue Data link layer issue

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

216. [UNKNOWN] unknown

75. A particle of mass 1 kg moves in a straight line under the influence of a force which increases linearly with time at the rate of 60 N/s, it being 40 N initially. The position of the particle after a lapse of 5 s, if it started from rest at the origin, will be 1250 m 1500 m 1750 m 2000 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

217. [UNKNOWN] unknown

76. Which of the following assumptions is/are made in the analysis of hydraulic jump? It is assumed that before and after jump formation the flow is essentially two-dimensional and that the pressure distribution is hydrostatic. The length of the jump is small so that the losses due to friction on the channel floor are small and hence neglected. The channel floor is horizontal or the slope is so gentle that the weight component of the water mass comprising the jump is very high. I only 2 only 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

218. [UNKNOWN] unknown

76. The Pre-echo PE distortions in audio signal represents the Theoretical limit on compressibility of particular signals Imaginary components of a signal Critical band analysis of a signal Histogram of the signals

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

219. [UNKNOWN] unknown

76. Rails are laid such that there will be no stress in them at 24 °C. If the rails are 32 m long with an expansion allowance of 8 mm per rail, coefficient of linear expansion $\alpha = 11 \times 10^{-6} / ^\circ\text{C}$ and $E = 205 \text{ GPa}$, the stress in the rails at 80 °C will be nearly 68 MPa 75 MPa 83 MPa 90 MPa A B C 101014—I LtNr4.3 kN lCN 14C-2 >K1 nig 3 m-01 If $E = 205 \text{ GPa}$, the total elongation of the bar will be nearly 29.7 mm 25.6 mm 2F5 mm 174 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

220. [UNKNOWN] unknown

77. Water is to be pumped out of a deep well under a total head of 95 m. A number of identical pumps of design speed 1000 rpm and specific speed 900 rpm with a rated capacity of 150 Us are available. The number of pumps required will be 1 3 5 7

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

221. [UNKNOWN] unknown

77. In a computer network, a point-to-point transmission, with one sender and one receiver is called Unicasting Multicasting Broadcasting Internetworkdng

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

222. [UNKNOWN] unknown

77. The loads acting on a 3 mm diameter bar at different points are as shown in the figure : (a) $f = 2n$, $2nNT$ f -

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

223. [UNKNOWN] unknown

78. Consider the following data from a test on Pelton wheel: Head at the base of the nozzle = 32 m Discharge of the nozzle = 018 m³/s Area of the jet = 7500 mm² Power available at the shaft = 44 kW Mechanical efficiency = 94% The power lost in the nozzle will be nearly 3.9 kW 4-7 kW 3.5 kW 2.3 kW k — 1 w2 2g VSY-S-CVL (13 — A) ===== PAGE 14 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

224. [UNKNOWN] unknown

78. The Protocol (http), the DNS name of the host, and the file name is identified through Uniform Resource Locator Web Browser Web Server IP address

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

225. [UNKNOWN] unknown

78. A hollow circular bar used as a beam has its outer diameter thrice the inside diameter. It is subjected to a maximum bending moment of 60 MN m. If the permissible bending stress is limited to 120 MPa, the inside diameter of the beam will be 49.2 mm 534 mm 57.6 mm 61.8 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

226. [UNKNOWN] unknown

79. A certain hydropower plant utilizes the flow as it occurs, without any provision for storage. It is premised that a defined minimum dry weather flow is available. Such a plant is classified as Diverted-flow plant Pooled storage plant Base-load plant Run-of-river plant

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

227. [UNKNOWN] unknown

79. The traditional way to handle forms and other interactive Web pages is a system called Graphical User Interface Common Gateway Interface Text Based User Interface Command Line Interface

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

228. [UNKNOWN] unknown

79. In a beam of I-section, which of the following parts will take the maximum shear stress when subjected to traverse loading? Flange Web Select the correct answer using the code given below. (a) 1 only (2.) 2 only Both 1 and 2 Neither 1 nor 2 VBY-S-MCH/50A 14 ===== PAGE 15 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

229. [UNKNOWN] unknown

80. Two turbo-generators, each of capacity 25,000 kW, have been installed at a hydel power station. The load on the hydel plant varies from 15,000 kW to 40,000 kW. The total installed plant capacity and the load factor are nearly 40,000 kW and 68-8% 50,000 kW and 68-8% 40,000 kW and 62.3% 50,000 kW and 62-3%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

230. [UNKNOWN] unknown

80. Pretty Good Privacy (PGP) which encrypts the data by using a block cipher is used in FTP security e-mail security Browser security Bluetooth security

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

231. [UNKNOWN] unknown

80. Which of the following statements is/are correct? In uniformly distributed load, the nature of shear force is linear and bending moment is parabolic. In uniformly varying load, the nature of shear force is linear and bending moment is parabolic. Under no loading condition, the nature of shear force is linear and bending moment is constant. Select the correct answer using the code given below. 1 and 2 1 and 3 2 only 1 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

232. [UNKNOWN] unknown

81. An airfoil is a streamlined body as shown in the figure below. Because of the streamlining of the body, the separation occurs only at the extreme rear of the body, resulting in

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

233. [UNKNOWN] unknown

81. The core elements of high-level programming languages are Keywords, Expressions and Punctuation Functions, Keywords and Operators Keywords, Operators and Punctuation Functions, Expressions and Operators 15 A — VBY-G-DST ===== PAGE 16 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

234. [UNKNOWN] unknown

81. The cross-section of the beam is as shown in the figure : H-200 mm---)11 110 mm 400 mm -710 mm 8 mm le 200 mm->1 If the permissible stress is 150 N/m/n2 , the bending moment M will be nearly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

235. [UNKNOWN] unknown

82. A plate 0.025 mm distant from a fixed plate moves at 60 cm/s and requires a force of 0.2 kg&m2 to maintain this speed. The dynamic viscosity of the fluid between the plates will be nearly 9-2 x 10-10 kgft/cm2 8-3 x 10-1° kgfs/cm2 7.4 x 10-10 kgfs/cm2 6-5 x 10-10 kgfs/cm2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

236. [UNKNOWN] unknown

82. The philosophical study of beliefs and knowledge is better known as Ontology Epistemology Entomology Etymology

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

237. [UNKNOWN] unknown

82. In a propeller shaft, sometimes apart from bending and twisting, end thrust will also develop stresses which would

be tensile in nature and uniform over the cross-section compressive in nature and uniform over the cross-section tensile in nature and non-uniform over the cross-section compressive in nature and non- uniform over the cross-section

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

238. [UNKNOWN] unknown

83. Which of the following are component parts for an oil pressure governor in modern turbines? Servomotor, known as relay cylinder Oil sump Oil pump which is driven by belt connected to turbine main shaft

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

239. [UNKNOWN] unknown

83. One branch of ethical philosophy claims that it is possible to know right from wrong or good from bad in a very clear and objective manner, is called Non-Cognitivism Ethical Pluralism Cognitivism Utilitarianism

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

240. [UNKNOWN] unknown

83. A spherical shell of 1.2 m internal diameter and 6 mm thickness is filled with water under pressure until volume is increased by $400 \times 10^3 \text{ mm}^3$. If $E = 204 \text{ GPa}$, Poisson's ratio $\nu = 0.3$, neglecting radial stresses, the hoop stress developed in the shell will be nearly 43 MPa 38 MPa 33 MPa 28 MPa

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

241. [UNKNOWN] unknown

84. A double-acting reciprocating pump having piston area 0.1 m^2 has a stroke 0.30 m long. The pump is discharging 2.4 m^3 of water per minute at 45 rpm through a height of 10 m. The slip of the pump and power required to drive the pump will be nearly $0.005 \text{ m}^3/\text{s}$ and 4.8 kW $0.003 \text{ m}^3/\text{s}$ and 4.8 kW $0.005 \text{ m}^3/\text{s}$ and 4.4 kW $0.003 \text{ m}^3/\text{s}$ and 4.4 kW A very high pressure drag A small wake and consequently small pressure drag A moderate pressure drag No pressure drag VBY-S-CVL (14 - A) ===== PAGE 15 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

242. [UNKNOWN] unknown

84. Consider the following statements regarding 'Engineering Ethics' : It is the activity of understanding moral values It resolves the moral issues and justifies moral judgments It would refer to the set of specifically moral problems and issues related to Engineering Which of the above statements are correct?

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

243. [UNKNOWN] unknown

84. The inner diameter of a cylindrical tank for liquefied gas is 250 mm. The gas pressure is limited to 15 MPa. The tank is made of plain carbon steel with ultimate tensile strength of 340 N/mm^2 , Poisson's ratio of 0.27 and the factor of safety of 5. The thickness of the cylinder wall will be (a) $1.21 \times 10^8 \text{ N mm}$ (a) 60 nun (b) $142 \times 10^8 \text{ N mm}$ 50 nun $1.64 \times 10^8 \text{ N mm}$ (c) 40 nun (d) $1.88 \times 10^8 \text{ N mm}$ (d) 30 nun VBY-S-MCH/ 50A 15 [P.T.O. ===== PAGE 16 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

244. [UNKNOWN] unknown

85. In intensity-duration analysis by Sherman, the intensity of rainfall i is represented as $b_n(t + a)$ (b)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

245. [UNKNOWN] unknown

85. A situation where very high prices are charged from customers for a limited period of time is known as Gouging Zipping Bamboozing Hoodwinking

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

246. [UNKNOWN] unknown

85. The structure of sodium chloride is

247. [UNKNOWN] unknown

86. Which one of the following points should be kept in mind while selecting the site for a rain gauge station? The site where a rain gauge is set up should be close to a meteorological observatory. The rain gauge should be on the top of a hill. A fence, if erected to protect the rain gauge from cattle etc. should be located within twice the height of the fence. The distance between the rain gauge and the nearest object should be at least twice the height of the object.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

248. [UNKNOWN] unknown

86. Consider the following steps for an individual regarding preparation for disclosure of unethical behaviour: Study and document the facts and formulate a plan for an appeal Take up the matter with higher management Discuss the matter with immediate supervisor If the internal appeal does not resolve the conflict, then he should notify the company that he intends to continue with an external review of the problem What is the correct sequence of order of the above steps? (a) 2, 3, 1 and 4 I, 2 and 3 1 and 2 only (b) 1, 3, 2 and 4 (c) 1 and 3 only (c) 3, 2, 4 and 1 (d) 2 and 3 only (d) 1, 2, 3 and 4
VBY-G-DST — A 16 ===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

249. [UNKNOWN] unknown

86. Hardenability of steel is assessed by Charpy impact test Rockwell hardness test Jominy end-quench test open-hole test

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

250. [UNKNOWN] unknown

87. Which of the following statements relates to a retarding reservoir? There are no gates at the outlets and hence the possibility of human error in reservoir operation is eliminated. The high cost of gate installation and also its operation is saved. An automatic regulation may cause coincidence of flood crest farther downstream where two or more channels taking off from retarding reservoirs join together. 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only Q es 2-72 Q /TS .
2-72

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

251. [UNKNOWN] unknown

87. Which of the following are the attributes of a profession? The work requires sophisticated skills, use of judgment and exercise of discretion Membership in the profession does not require extensive formal education as well as practical training There are set standards for admission to the profession and conduct for members Significant public good results from practice of the profession

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

252. [UNKNOWN] unknown

87. A metal has lattice parameter of 2-9 Å, density of 7-87 g/cc, atomic weight of 55-85, and Avogadro's number is 6.0238×10^{23} . The number of atoms per unit cell will be nearly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

253. [UNKNOWN] unknown

88. The coefficient of transmissibility T for a confined aquifer can be determined by a pumping-out test together with other relevant observations. The applicable formula is (where Q = Discharge, and Δs = Difference in drawdowns in two wells) (a) $\frac{2.72 AS}{T} \log \frac{r_2}{r_1}$ (b) $\frac{2.72 AS}{T} \log \frac{r_2}{r_1}$ (c) $\frac{2.72 AS}{T} \log \frac{r_2}{r_1}$ (d) $\frac{2.72 AS}{T} \log \frac{r_2}{r_1}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

254. [UNKNOWN] unknown

88. What are the core qualities of a professional practitioner? Integrity both with themselves and with others Independence-to be free of secondary interests with other parties Competence Discretion-care with communications

255. [UNKNOWN] unknown

88. considered as (a) a body-centered crystal An atomic packing factor (APF) for the BCC unit cell of hard spheres atoms will be (b) a simple cubic crystal (a) 0-63 (c) two interpenetrating FCC sub- lattices of Cl^- ions and Na^+ ions (b) 0-68 (d) a cubic crystal with Na^+ and Cl^- 0-73 alternatively at the cubic corners 0-78

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

256. [UNKNOWN] unknown

89. The volume of water below the minimum pool level in a reservoir is known as Useful storage Surcharge storage Dead storage Bank storage

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

257. [UNKNOWN] unknown

89. When should whistle blowing be attempted ? There must be a clear and great harm that can be avoided The whistleblower must be in a clear position to report on the problem The whistleblower must have a reasonable chance of success in stopping the harmful activity The whistleblower feels that all other lines of action within the context of the organization have been explored and shut off Select the correct answer using the codes given below: Select the correct answer using the codes given below: (a) 1,2 and 3 only (a) 1, 2, 3 and 4 (b) 1,2 and 4 only 1, 2 and 4 only 1,3 and 4 only (c) 1, 3 and 4 only (d) 2, 3 and 4 only (d) 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

258. [UNKNOWN] unknown

89. The distinct characteristic of Invar is it is magnetic it has low coefficient of thermal expansion it has high tensile strength it is non-corrosive

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

259. [UNKNOWN] unknown

90. Depending upon the source from which the water is drawn, flow irrigation can be 'sub-divided into River canal irrigation Reservoir or tank irrigation Combined storage and lift irrigation Combined storage and diversion irrigation Which of the above designations are relevant? 1, 2 and 3 only 1, 2 and 4 only 1, 3 and 4 only 2, 3 and 4 only
VBYS-CVL (15 — A) ===== PAGE 16 ===== 9.1.1, &insider the following data Root ione depth = 2 m Existing water content = 5% Dry density of soil = 15 kN/m³ Water applied to the soil = 500 mm³ Water loss due to evaporation and deep percolation = 10% Area of plot = 1000 m² The field capacity of the soil will be nearly 16.8% 17-7% 18-8% 19.7%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

260. [UNKNOWN] unknown

90. Which of the following are the salient features of the Patent Act 1970? It codifies inventions which are not patentable It provides for endorsement of patent with the words 'license of right' It provides for revocation of patents in public interest It has provision for validity period also for the patents Select the correct answer using the codes given below: 1, 2, 3 and 4 1, 2 and 3 only 1 and 3 only 3 and 4 only Select the correct answer using the codes given below: 1, 2, 3 and 4 1, 2 and 4 only 1, 3 and 4 only 2 and 3 only 17 A — V13Y-G-DST ===== PAGE 18 ===== Directions : Each of the next Ten (10) items consists of two statements, one labelled as the 'Statement al and the other as 'Statement (H)'. You are -to examine these two statements carefully and select the answers to these items using the codes given below: Codes: Both Statement (I) and Statement (H) are individually true and Statement (II) is the correct explanation of Statement (I) Both Statement (I) and Statement (II) are individually true but Statement (II) is not the correct explanation of Statement (I) Statement (I) is true but Statement (II) is false Statement (I) is false but Statement (II) is true.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

261. [UNKNOWN] unknown

90. An alloy produced by adding 1% of tin to Muntz metal is called as (a) 1 (a) a brass (b) 2 Admiralty brass 8 (c) Naval brass (d) 16 (d) Lead brass VBYS-MCH/50A 16 ===== PAGE 17 =====

262. [UNKNOWN] unknown

91. Statement (I) : All projects have constraints or limitations that inhibit their ability to reach goals and objectives. Statement (II) : Time and money are universal constraints in projects.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

263. [UNKNOWN] unknown

91. A sample of glass has a crack of half-length 2 mm. The Young's modulus of glass is 70 GN/m² and specific surface energy is 1 J/m². The fracture strength will be 885 MPa 895 MPa 915 MPa 935 MPa

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

264. [UNKNOWN] unknown

92. Consider the following data for irrigation water: Concentration Milli-equivalents per litre 1 Na + 24 Ca²⁺ 3-6 3' me+ 2 The Sodium-Absorption Ratio (SAR) is nearly (a) 13.1 14.3 15.5 16-7

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

265. [UNKNOWN] unknown

92. Statement (I): Training should be conducted among the line and low management for ensuring the importance of environmental protection plan. Statement (II) : Environmental science is a developing subject and the people implementing environment strategies should remain up to date with the environmental control processes. Statement (I) : Metals having same crystal structure will have greater solubility. Statement (II) : Differences in crystal structure limits the solid solubility. Statement (I) : The tie line is constructed across the two-phase region at the temperature of the alloy. Statement (II) : The overall alloy composition is located on the tie line. Statement (I) : Cross linked polymers may be synthesized in which side-branch chains are connected to the main ones. Statement (II) : Linear polymers are those in which the repeat units are joined together end to end in single chains. Statement (I) : Abrasive ceramics are used to wear, grind, or cut away other material, which necessarily is softer. Statement (II) : The prime requisite for abrasive ceramic group of materials is hardness or wear resistance and a high degree of toughness is essential to ensure that the abrasive particles do not easily fracture. VBY-G-DST — A 18 ===== PAGE 19 ===== Statement (I) : • The prevention costs increase with the introduction of a quality system and may be a significant proportion of the total quality costs. Statement (II) : Costs associated with education and training are not included in prevention costs. Statement (I) : An emulator is not a mixture of hardware and software and it cannot be used to test and debug the hardware and software of an external system. Statement (II) : Part of the hardware of an emulator is a multiwire cable which connects the host system to the system being developed. Statement (I) : Agency-loyalty is acting to fulfill one's contractual duties to an employer. Statement (II) : Agency-loyalty is entirely a matter of actions, whatever its motives. Statement (I) : An ETA is a study of the probable changes in socio-economic and biophysical characteristics of the environment that may result from a proposed action. Statement (II) : The purposes of an ETA is to help design projects, which do not disturb the quality of the environment by examining alternatives. 19 A — V13Y-G-DST

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/GENERAL-STUDIES-ENGINEERING-APTITUDE-ESEP-19_0.pdf

266. [UNKNOWN] unknown

92. In the Pb-Sn system, the fraction of total α phase is 3 times the fraction of β phase at eutectic temperature of 182 °C, Pb with 19% Sn dissolved in it, Sn with 2.5% Pb dissolved in it, and liquid is in equilibrium. The alloy compositions of tin (Sn) and lead (Pb) are nearly 28.6% and 71.4% 38.6% and 61.4% 48.6% and 51.4% 58.6% and 41.4%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

267. [UNKNOWN] unknown

93. A cylindrical specimen of steel having an original diameter of 12.8 mm is tensile tested to fracture and found to have engineering fracture strength σ_f of 460 MPa. If its cross-sectional diameter at fracture is 10.7 mm, the true stress at fracture will be 660 MPa 645 MPa 630 MPa 615 MPa

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

268. [UNKNOWN] unknown

94. Consider the following data while designing an expansion transition for a canal by Mitra's method: Length of flume = 16 m Width of throat = 9 m Width of canal = 15 m If B_x is the width at any distance x from the flumed section, the values of B_x at $x = 8$ m and at $x = 16$ m are nearly 10.8 m and 15 m 11.3 m and 15 m 10.8 m and 13 m 11.3 m and 13 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

269. [UNKNOWN] unknown

94. An iron container 10 cm x 10 cm at its base is filled to a height of 20 cm with a corrosive liquid. A current is produced as a result of an electrolytic cell, and after four weeks, the container has decreased in weight by 70 g. If it = 2, $F = 96500$ C and $M = 55-84$ g/mole, the current will be 0.05 A 0.10 A 0.20 A 0.40 A

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

270. [UNKNOWN] unknown

95. Consider the following data for a drain: $L = 50$ m, $a = 10$ m, $b = 10-3$ m, and $k = 1 \times 10^{-5}$ m/s If the drains carry 1% of average annual rainfall in 24 hrs, the average annual rainfall for which this system has been designed will be 78 cm 84 cm 90 cm 96 cm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

271. [UNKNOWN] unknown

95. A copper piece originally 305 mm long is pulled in tension with a stress of 276 MPa. If the deformation is entirely elastic and the modulus of elasticity is 110 GPa, the resultant elongation will be nearly 0.43 mm 0.54 mm 0.65 mm 0.77 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

272. [UNKNOWN] unknown

96. The purpose of constructing a 'Groyne' is to Expand a river channel to improve its depth Encourage meandering Train the flow along a certain course Reduce the silting in the river bed

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

273. [UNKNOWN] unknown

96. The indentation on a steel sample has been taken using 10 mm tungsten carbide ball at 500 kgf load. If the average diameter of the indentation is 2.5 mm, the BHN will be nearly 90 100 110 120 VBY-S-MCH/ 50A 17 [P.T.O. =====
PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

274. [UNKNOWN] unknown

97. Which one of the following compounds of nitrogen, when in excessive amounts in water, contributes to the illness known as infant methemoglobinemia ? Ammoniacal nitrogen Albuminoid nitrogen Nitrite Nitrate VBY-S-CVL (16 —'A),
===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

275. [UNKNOWN] unknown

97. Which of the following statements are correct with respect to inversion of mechanisms? It is a method of obtaining different mechanisms by fixing different links of the same kinematic chain. It is a method of obtaining different mechanisms by fixing the same links of different kinematic chains. In the process of inversion, the relative motions of the links of the mechanisms produced remain unchanged. In the process of inversion, the relative motions of the links of the mechanisms produced will change accordingly. Select the correct answer using the code given below.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

276. [UNKNOWN] unknown

98. Consider the following data regarding a theoretical profile of a dam: Permissible value of compressible stress $a = 350$ tonnes/m² Specific gravity of concrete $s = 2.4$ Uplift coefficient $c = 0.6$ m The value of $y = 1$ The height and base

width will be nearly 125 m and 63 m 175 m and 63 m 125 m and 93 m 175 m and 93 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

277. [UNKNOWN] unknown

98. For the follower with stroke S , following the cycloidal motion, the radius of the rolling circle will be (a) $S \times 2\pi$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

278. [UNKNOWN] unknown

99. Consider the following statements with respect to weir under discussion:

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

279. [UNKNOWN] unknown

99. Chlorine usage in the treatment of 25,000 m³/day of water has been 9 kg/day. The residual chlorine after 10 minutes contact is 0.2 mg/l. The chlorine demand of water would be nearly 0.28 mg/l 0.22 mg/l 0.16 mg/l 0.12 mg/l

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

280. [UNKNOWN] unknown

99. A vertical shaft of 100 mm diameter and 1 m length has its upper end fixed at the top. The other end carries a disc of 5000 N and the modulus of elasticity of the shaft material is 2×10^5 N/mm². Neglecting the weight of the shaft, the frequency of the longitudinal vibrations will be nearly 279.5 Hz 266.5 Hz 253.5 Hz 241.5 Hz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

281. [UNKNOWN] unknown

100. The demand of water is 150 litres/head/day in a city of one lakh population. The factor of safety is taken as 1.5, detention time as 4 h and overflow rate as 20,000 litres/day/m². The area of 3 m deep plain sedimentation tank as per surface loading consideration will be 1025 m² 1075 m² 1125 m² 1175 m²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

282. [UNKNOWN] unknown

100. The accurate method of finding the natural frequency of transverse vibrations of a system of several loads attached to some shaft is (a) 1 and 3 (a) Dunkerley method (b) 1 and 4 energy method 2 and 3 (c) Stodola method (d) 2 and 4 (d) Dunkerley and energy method

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

283. [UNKNOWN] unknown

101. The rain intensity over 54 hectares of land is 50 mm/h, 30% of area consists of roof surfaces with runoff rate as 0.9, 30% is open field with runoff rate of 0.2 and remaining 40% is road network with runoff rate of 0.4. The storm water flow will be nearly 2.6 m³/s 3.7 m³/s 4.8 m³/s 5.9 m³/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

284. [UNKNOWN] unknown

101. The speed at which the shaft runs, so that the deflection of the shaft from the axis of rotation becomes infinite, is known as whipping speed damping speed resonant speed gravitational speed $2n \rightarrow 2n \sqrt{S + 27-c}$ VBY-S-MCH/50A 18

===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

285. [UNKNOWN] unknown

102. Critical dissolved oxygen (D.O.) deficit occurs in which one of the following zones of pollution of 'oxygen sag curve' in case of self-purification of natural streams? Zone of recovery Zone of active decomposition Zone of degradation Zone of clear water

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

286. [UNKNOWN] unknown

102. Which one of the following is not the correct statement with respect to the involute profile toothed gears in mesh? Pressure angle remains constant from the start till the end of the engagement. The base circle diameter and the pitch circle diameter of the two mating involutes are proportional. When two involutes are in mesh, the angular velocity ratio is proportional to the size of the base circles. The shape of the involute profile depends only on the dimensions of the base circle.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

287. [UNKNOWN] unknown

103. The MLSS concentration in an aeration tank is 2000 mg/l and the sludge volume after 30 minutes of settling in a 1000 ml graduated cylinder is 176 ml. The value of sludge density index (SDI) will be nearly 3-34 g/ml 2.22 g/ml 1.14 g/ml 0.26 g/ml

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

288. [UNKNOWN] unknown

103. The centre distance C between two gears, in terms of base circle radii R_1 , R_2 and the pressure angle ϕ , is c (a) $0.5(R_1 + R_2) \cos \phi$ (b) $0.5(R_1 + R_2) \sin \phi$ (c) $0.5(R_1 + R_2) \tan \phi$ (d) $0.5(R_1 + R_2) \sec \phi$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

289. [UNKNOWN] unknown

104. Which one of the following gases is the principal by-product of anaerobic decomposition of the organic content in waste water? Carbon monoxide Ammonia Hydrogen sulphide Methane

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

290. [UNKNOWN] unknown

104. A three-cylinder single-acting engine has its cranks at 120° . The turning moment diagram for each cycle is a triangle for the power stroke with a maximum torque of 60 N m at 60° after the dead centre of the corresponding crank. There is no torque on the return stroke. The engine runs at 400 r.p.m. The power developed will be 1745 W 1885 W 1935 W 1995 W

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

291. [UNKNOWN] unknown

105. Consider the following statements with reference to the mixing of industrial waste water with domestic waste water: The industrial waste water can be mixed with domestic water when it has higher BOD. The industrial waste water can be mixed with domestic water when the pH value of industrial waste water is highly alkaline. Which of the above statements is/are correct? 1 only 2 only Both 1 and 2 Neither 1 nor 2 VBY-S-CVL (17 — A) ===== PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

292. [UNKNOWN] unknown

105. A vertical single-cylinder opposed piston type engine has reciprocating parts of mass 2000 kg for the lower piston and 2750 kg for the upper piston. The lower piston has a stroke of 60 cm and the engine is in primary balance. If the ratio of the length of connecting rod to crank is 4 for the lower piston and 8 for the upper piston, and when the crankshaft speed is of 135 r.p.m., the maximum secondary unbalanced force will be 48935.5 N 46946.5 N 44968.5 N 42989.5 N VBY-S-MCH/ 50A 19 [P.T.O. ===== PAGE 20 ===== —150 .mm 120 mm 180 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

293. [UNKNOWN] unknown

106. The waste water from a factory having a pH of 10, contains KOH only For waste water discharge is 80 m³/day. The total quantity of KOH per day will be nearly 4-5 kg/day 5.4 kg/day 61 kg/day (ft) 72 kg/day

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

294. [UNKNOWN] unknown

106. The reciprocating mass is balanced when primary force is balanced by the mass = $cmr [1]2 \cos 9$ unbalanced by the mass = $cmr \cos 2 \cos e$ balanced by the mass = $(1- c) cmr (02 \cos 9$ unbalanced by the mass = $(1- c)cmr \cos 2 \cos 9$
Select the correct answer using the code given below. 1 and 3 2 and 3 1 and 4 2 and 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

295. [UNKNOWN] unknown

107. Fanning type of plume behaviour takes place when Super-adiabatic lapse rate prevails with light to moderate wind speed Extreme inversion conditions exist in the presence of light wind There exists a strong super-adiabatic lapse rate above a surface of inversion Plume is caught between two inversion layers

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

296. [UNKNOWN] unknown

107. The active gyroscopic torque in gyroscope about a horizontal axis represents the torque required to cause the axis of spin to precess in the vertical plane the torque required to cause the axis of spin to precess in the horizontal plane

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

297. [UNKNOWN] unknown

108. A thermal power plant burns coal at the rate of. 8 t/h. The coal has sulphur content of 4-5%. The rate of emission of SO₂ will be 180 g/s 200 g/s 220 g/s 240 g/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

298. [UNKNOWN] unknown

108. The change in governor height for a Watt governor when speed varies from 100 r.p.m. to 101 r.p.m. will be nearly 1.8 mm 2.6 mm 3.4 mm 4.2 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

299. [UNKNOWN] unknown

109. The property, of clays by virtue of which they regain, if left alone for a time, a part of the strength lost due to remoulding at unaltered moisture content, is known as Thixotropy Sensitivity Consistency Activity VBY-S-CVL

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

300. [UNKNOWN] unknown

109. A rectangular strut is 150 mm wide and 120 mm thick. It carries a load of 180 kN at an eccentricity of 10 mm in a plane bisecting the thickness as shown in the figure : The maximum intensity of stress in the section will be the force required to cause the axis (a) 14 MPa of spin to precess in the horizontal plane (b) 12 MPa the force required to cause the axis (c) 10 MPa of spin to precess in the vertical plane (d) 8 MPa VBY-S-MCH/50A 20 ===== PAGE 21 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

301. [UNKNOWN] unknown

110. The plastic limit and liquid limit of a soil are 30% and 42% respectively. The percentage -Volume change from liquid limit to dry state is 35% of the dry volume. Similarly the percentage volume change from plastic limit to dry state is 22% of the dry volume. The shrinkage ratio will be nearly 4-2 3.1, 2-2 1.1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

302. [UNKNOWN] unknown

110. The theory of failure used in designing the ductile materials in a most accurate way is by maximum principal stress theory distortion energy theory maximum strain theory Select the correct answer using the code given below. 1, 2 and 3 1 only 2 only 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

303. [UNKNOWN] unknown

111. The ratio of a given volume change in a soil, expressed as percentage of the dry volume, to the corresponding change in water content is called (a) Specific gravity of soil solids (h) Mass-specific gravity of soils Shrinkage ratio of soils Density ratio of soils

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

304. [UNKNOWN] unknown

111. When a load of 20 k.N is gradually applied at a particular point in a beam, it produces a maximum bending stress of 20 MPa and a deflection of 10 mm. What will be the height from which a load of 5 l.cN should fall onto the beam at the same point if the maximum bending stress is 40 MPa? 80 nun 70 nun 60 mm 50 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

305. [UNKNOWN] unknown

112. A masonry dam is founded on pervious sand. A factor of safety of 4 is required against boiling. For the sand, $n = 45\%$ and $G_s = 2.65$. The maximum permissible upward hydraulic gradient will be nearly 0.18 0.23 0.28 0.33

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

306. [UNKNOWN] unknown

112. The areas of fatigue failure in a part may be in the region having slow growth of crack with a fine fibrous appearance. region having faster growth of crack with a fine fibrous appearance. region of sudden fracture with a coarse granular appearance. region of gradual fracture with a coarse granular appearance. Select the correct answer using the code given below. 2 and 3 2 and 4 1 and 4 1 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

307. [UNKNOWN] unknown

113. The representative liquid limit and plastic limit values of a saturated consolidated clay deposit are 60% and 30%, respectively. The saturated unit weight of the soil is 19 kN/m³. The water table is at 8 m below ground level. At a depth of 10 m from the ground surface, the undrained shear strength of the soil will be nearly 37.7 kN/m² 33.5 kN/m² 29.3 kN/m² 25.1 kN/m² (18—A). ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

308. [UNKNOWN] unknown

113. The shock-absorbing capacity (resilience) of bolts can be increased by increasing the shank diameter above the core diameter of threads reducing the shank diameter to the core diameter of threads decreasing the length of shank portion of the bolt pre-heating of the shank portion of the bolt

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

309. [UNKNOWN] unknown

114. A 6 m high retaining wall with a vertical back has a backfill of silty sand with a slope of 100 for the backfill. With values of $K_{II} = 760 \text{ kg/m}^2/\text{m}$ and $K_v = 100 \text{ kg/m}^2/\text{m}$, the total active earth pressure will approximately be 128 kN/m 134 kN/m 138 kN/m 142 kN/m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

310. [UNKNOWN] unknown

114. The torque required to tighten the bolt comprises of the torque required in overcoming thread friction only torque required in inducing the pre-load only torque required in overcoming circumferential hoop stress torque required in overcoming thread friction and inducing the pre-load and also the torque required to overcome collar friction between the nut and the washer VBY-S-MCH/50A 21 P.T.O. ===== PAGE 22 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

311. [UNKNOWN] unknown

115. The vertical stress at any point at a radial distance r and at depth z as determined by using Boussinesq's influence

factor KB and Westergaard's influence factor Kw would be (almost same for —r ratios equal to or greater z than 2-0 1-8 1-5 1-2

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

312. [UNKNOWN] unknown

115. A steel spindle transmits 4 kW at 800 r.p.m. The angular deflection should not exceed 0.25°/m length of the spindle. If the modulus of rigidity for the material of the spindle is 84 GPa, the diameter of the spindle will be 46 mm 42 mm 38 mm 34 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

313. [UNKNOWN] unknown

116. A strip footing 2 m in width, with its base at a depth of 1-5 m below ground surface, rests on a saturated clay soil with $\gamma_{sat} = 20 \text{ kN/m}^3$; $C_u = 40 \text{ kN/m}^2$; $\phi = 0$; $c' = 10 \text{ kN/m}^2$; and $\alpha = 20^\circ$. The natural water table is at 1 m depth below ground level. As per IS: 6403 — 1981, the ultimate bearing capacity of this footing will be (taking the relevant N_s as 5-14)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

314. [UNKNOWN] unknown

116. A taper roller bearing has a dynamic load capacity of 26 kN. The desired life for 90% of the bearings is 8000 hr and the speed is 300 r.p.m. The equivalent radial load that the bearing can carry will be nearly 5854 N 5645 N 5436 N 5227 N

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

315. [UNKNOWN] unknown

117. The settlement due to secondary compression is predominant in Granular soils Inorganic clays Organic clays Very fine sand and silts

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

316. [UNKNOWN] unknown

117. Hollow shafts are stronger than solid shafts having same weight because the stiffness of hollow shaft is less than that of solid shaft the strength of hollow shaft is more than that of solid shaft the natural frequency of hollow shaft is less than that of solid shaft in hollow shafts, material is not spread at large radius

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

317. [UNKNOWN] unknown

118. A raft foundation 10 m wide and 12 m long is to be constructed in a clayey soil having shear strength of 12 kN/m². Unit weight of soil is 16 kN/m³. The ground surface carries a surcharge of 20 kN/m²; the factor of safety is 1-2 and the value of $N_s = 5-7$. The safe depth of foundation will be nearly 8-2 m 7-3m 6-4m 5-5 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

318. [UNKNOWN] unknown

118. A propeller shaft is required to transmit 45 kW power at 500 r.p.m. It is a hollow shaft having inside diameter 0.6 times the outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm². The inner and outer diameters of the shaft are nearly 21.7 mm and 391 mm 23-5 mm and 391 mm 21.7 mm and 32.2 mm 23-5 mm and 32.2 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

319. [UNKNOWN] unknown

119. The skin frictional resistance of a pile driven in sand does not depend on Lateral earth pressure coefficient Angle of friction between pile and soil Pile material Total stress analysis

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

320. [UNKNOWN] unknown

119. A bicycle and rider travelling at 12 km/hr on a level road have a mass of 105 kg. A brake is applied to a rear wheel having 800 mm diameter. The pressure on the brake is 80 N and the coefficient of friction is 0.06. The number of turns of the wheel before coming to rest will be 48.3 revolutions 42.6 revolutions 38.3 revolutions 32.6 revolutions

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

321. [UNKNOWN] unknown

120. An excavation is made with a vertical face in a clay soil which has $C_u = 50 \text{ kN/m}^2$, $\gamma_t = 18 \text{ kN/m}^3$ and $s_n = 0.261$. The maximum depth of a stable excavation will be nearly (19 — A) 10-6 m 12-4th 14-2 m 16-0 m 327 kN/m² 285 kN/m² 253 kN/m² 231 kN/m² VBYS-CVL ===== PAGE 20 ===== Reconnaissance survey for determining feasibility and estimation of scheme falls under the classification based on the Nature of the field of survey Object of surveying Instruments used Method employed A survey line BAC crosses a river, A and C being on the near and distant bank respectively. Standing at D, a point 50 m measured perpendicularly to AB from A, the bearings of C and B are 320° and 230° respectively, AB being 25 m. The width of the river will be 80 m 90 m 100 m 110 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

322. [UNKNOWN] unknown

120. To avoid self-engagement in cone clutch, its semi-cone angle is always kept smaller than the angle of static friction equal to the angle of static friction greater than the angle of static friction half of the angle of static friction VBYS-MCH/ 50A 22 ===== PAGE 23 ===== The average roughness R_a for sampling length of 0.8 mm will be 1.14 μm 1.10 μm 1.06 μm F02 μm r 500 r 450 r 400 r 350

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

323. [UNKNOWN] unknown

121. In case of arc welding of steel with a potential of 20 V and current of 200 A, the travel speed is 5 mm/s and the cross-sectional area of the joint is 20 mm². The heat required for melting steel may be taken as 10 J/mm³ and heat transfer efficiency as 0.85. The melting efficiency will be nearly 18% 29% 36% 42%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

324. [UNKNOWN] unknown

122. What is the force required for 90° bending of St50 steel of 2 mm thickness in a V-die, if the die opening is taken as 8 times the thickness and the length of the bent part is 1 m, ultimate tensile strength is 500 MPa and $K = 1.33$? 166.25 kN 155.45 kN 154.65 kN 143.85 kN

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

325. [UNKNOWN] unknown

123. In plane surveying where a graduated staff is observed either with horizontal line of sight or inclined line of sight, the effect of refraction is to Increase the staff reading Decrease the staff reading Neither increase nor decrease the staff reading Duplicate the staff reading

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

326. [UNKNOWN] unknown

123. A graph is drawn to a vertical magnification of 10000 and horizontal magnification of 100, and the areas above and below the datum line are as follows : Above 150 mm² 80 mm² 170 mm² 40 mm² Below 80 mm² 60 mm² 150 mm² 120 mm²

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

327. [UNKNOWN] unknown

124. A sidereal day is the average time taken by The Earth to move around the Sun once The Moon to move around the Earth once The first point of Aries to cross the same meridian successively The Earth to move around its own axis once

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

328. [UNKNOWN] unknown

124. The radius of arc is measured by allowing a 20 mm diameter roller to oscillate to and fro on it and the time for 25 oscillations is noted at 56-25 s. The radius of arc will be 865 mm 850 mm 835 mm 820 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

329. [UNKNOWN] unknown

125. In triangulation, in order to control the accumulation of errors of length and azimuth subsidiary bases are selected. At certain stations, the astronomical observations for azimuth and longitude are also made. These stations are called Transportation stations Bowditch stations Universe stations Laplace stations

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

330. [UNKNOWN] unknown

125. Which one of the following systems is consisting of processing stations, material handling and storage, computer control system and human labour? Portable manufacturing system Focused integrated system Flexible manufacturing system Automated integrated system

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

331. [UNKNOWN] unknown

126. A vertical photograph is taken at an altitude of 1200 m 'above mean sea level' (a.m.s./.) of a terrain lying at an elevation of 80 m a.m.s./. The focal length of camera is 15 cm. The scale of the photograph will be nearly

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

332. [UNKNOWN] unknown

126. A project initially costs ₹ 5,000 and generates year-end cash inflows of ₹ 1,800, ₹ 1,600, ₹ 1,400, ₹ 1,200 and ₹ 1,000 respectively in five years of its life. If the rate of return is 10%, the net present value (NPV) will be ₹ 50A 23 [P.T.O. ===== PAGE 24 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

333. [UNKNOWN] unknown

127. Aerial photographs are required to be taken to cover an area of 150 km². The longitudinal and side overlaps are to be 60% and 30% respectively. The scale of photograph is 1 cm = 100 m; and the size of each photograph is 20 cm x 20 cm. The minimum required number of photographs will be 170 158 146 134 VBY-S-CVL (20—A) ===== PAGE 21 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

334. [UNKNOWN] unknown

127. What is the mode for the following distribution? Gross profit as percentage of sales Number of companies 0-7 19 7-14 25 14-21 36 21-28 72 28-35 51 35-42 43 42-49 28 49-55 21 55-60 23 60-65 25 65-70 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

335. [UNKNOWN] unknown

128. Which one of the following conditions is correct with respect to the transition curve? It should be tangential to the straight approaches at the two ends. It should meet the circular curve tangentially. Its curvature will necessarily be non-zero at the point of take-off from the straight approaches. The rate of increase of curvature along the transition reach should match with the increase of cant.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

336. [UNKNOWN] unknown

128. Consider the following data for quality acceptance process : N= 10000 n = 89 c = 2 p = 0.01 (incoming lots of quality) Pa = 0.9397

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

337. [UNKNOWN] unknown

129. A circular curve has a long chord of 80 m and a versed sine of 4 m. The height and ordinate at a distance of 30 m from the mid-ordinate will be nearly 306m 2-72m 224m 1.76 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

338. [UNKNOWN] unknown

129. An engine is to be designed to have a minimum reliability of 0-8 and minimum availability of 0-98 over a period of 2 x 10³ hr. The MTTR is nearly 168 hr 174 hr 183 hr 188 hr

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

339. [UNKNOWN] unknown

130. Two parallel railway lines are to be connected by a reverse curve, each section having the same radius. If the lines are 12 m apart and the maximum distance between tangent points measured parallel to the straights is 48 m, then the maximum allowable radius will be 511m 523m 535m 54-7 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

340. [UNKNOWN] unknown

130. Which one of the following relations with usual notations will hold good in a dynamic vibration absorber system under tuned conditions? $k_1 l_2 = m_1 r^2$ $k_1 m_2 = m_1 k_2$ $k_1 m_1 = k_2 m_2$ $k_1 + k_2 = + m_2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

341. [UNKNOWN] unknown

131. In an old map, a line AB was drawn to a magnetic bearing of 5°30', the magnetic declination at the time being 1° East. If the present magnetic declination is 8°30' East, the line should be set to a magnetic bearing of 358° 2° 6°30' 357°

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

342. [UNKNOWN] unknown

131. In ultrasonic waves, the frequencies for non-destructive testing of materials are in the range of The AOQ will be (a) 0-5 MHz to 10 MHz (a) 0-93% (b) 10 MHz to 20 MHz 0-84% 20 MHz to 30 MHz (c) 0-75% (d) 0-66% (d) 30 MHz to 40 MHz VBY-S-MCH/ 50A 24 ===== PAGE 25 ===== battery-backed relay dummy relay one-shot operation (d) master control relay 3.4 m³/s 3.8 m³/s 4.2 m³/s 46 m³/s

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

343. [UNKNOWN] unknown

132. An unconformity is A surface of erosion or non-deposition as detected in a sequence of rocks A layer of boulders and pebbles in a sequence of rocks A layer of clay or shale in an igneous mass A type of joint especially associated with folded and faulted rocks

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

344. [UNKNOWN] unknown

133. Consider two cars approaching from the opposite directions at 90 km/h and 60 km/h. If the reaction time is 2.5 s, coefficient of friction is 0.7 and brake efficiency is 50% in both the cases, the minimum sight distance required to avoid a head-on collision will be nearly 154 m 188 m 212 m 236 m VBY-S-CVL 21 — A) ===== PAGE 22 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

345. [UNKNOWN] unknown

133. Which of the following is one of the basic units of memory controller in micro- controller? (a) Microcode engine (14 Master program counter Program status word Slave program counter

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

346. [UNKNOWN] unknown

134. What is the extra widening required (as nearest magnitude) for a pavement of 7 m width on a horizontal curve of

radius 200 m, if the longest wheel of vehicle expected on the road is 6.5 m and the design speed is 65 km/h ? 0-3 m
0.5m , 0.7 m 0.9 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

347. [UNKNOWN] unknown

134. Which one of the following ways will be adopted to store the program counter contents? Last-in-First-out (LIFO)
First-in-First-out (FIFO) Last-in-Last-out (LILO) First-in-Last-out (FILO)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

348. [UNKNOWN] unknown

135. A vehicle moving at 40 km/h speed was stopped by applying brake and the length of the skid mark was 12.2 m. If the average skid resistance of the pavement is 0.70, the brake efficiency of the test vehicle will be nearly 80% 74% 68% 62%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

349. [UNKNOWN] unknown

135. In ladder logic programming, an alternative in place of using same internal relay contact for every rung is to use The term 'attenuation' is used to describe the process of removing a certain band of frequencies from a signal and permitting others to be transmitted. The Wheatstone bridge can be used to convert a voltage change to an electrical resistance change. Which of the above statements is/are correct? 1 only 2 only Both 1 and 2 Neither 1 nor 2 At time t, the excitation voltage to a resolver is 24 V. The shaft angle is 90°. The output signals from the resolver V_{s1} and V_{s2} will be 12 V and 0 V 24 V and 0 V 12 V and 12 V 24 V and 12 V An actuator having a stem movement at full travel of 30 mm is mounted with a control valve having an equal percentage plug and with minimum flow rate of 2 m³/s and maximum flow rate of 24 m³/s. When the stem movement is 10 mm, the flow rate will be VBY-S-MCH/ 50A 25 [P.T.O. =====
PAGE 26 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

350. [UNKNOWN] unknown

136. The main drawback of automatic counters-cum-classifiers, used for traffic volume studies, is that it is not yet possible to classify and record Vehicle type Axle spacing Axle load Speed

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

351. [UNKNOWN] unknown

136. Consider the following statements : magnetic materials is about 390 °C 540 °C 760 °C 880 °C

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

352. [UNKNOWN] unknown

137. Which one of the following is not a part of 'speed and delay' studies? Floating car method Vehicle number method Interview technique License number method

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

353. [UNKNOWN] unknown

138. Consider the following data with respect to the design of flexible pavement: Design wheel load = 4200 kg Tyre pressure = 6.0 kg/cm² Elastic modulus = 150 kg/cm² Permissible deflection = 0.25 cm (take TC₁₁₂ = 1-77, 7C₁₁₂ = 0-564, = 0-318, r_c and i_{c2} = 9.87) The total thickness of flexible pavement for a single layer elastic theory will be nearly 42 cm 47 cm 51 cm 56 cm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

354. [UNKNOWN] unknown

139. The minimum possible grade that can be provided in a tunnel and its approaches with providing adequately for proper drainage is 0.1% C/2% 0.3% 0.4% VBY-S-CVL (22 — A) ===== PAGE 23 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

355. [UNKNOWN] unknown

139. In a rack and pinion system, rack is an element moving in translational direction and pinion is a rotary gear. Which one of the following statements is correct? (a) Translational acceleration is directly proportional to the moment of inertia of pinion. (b) Translational acceleration is inversely proportional to the moment of inertia of pinion. Angular acceleration is inversely proportional to the torque on pinion shaft. Translational velocity is directly proportional to the moment of inertia of pinion.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

356. [UNKNOWN] unknown

140. The section of the tunnel adopted perfectly in lieu of ease of construction and maintenance in hard rock tunnels, where the risk of roof failure or collapse caused by external pressure from water, or from loose or unstable soil conditions on tunnel lining is practically non-existent, is Circular section Segmental roof section Horse-shoe section Egg-shaped section

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

357. [UNKNOWN] unknown

140. For the control signal to change at a rate proportional to the error signal, the robotic controller must employ (a) integral control

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

358. [UNKNOWN] unknown

141. Which one of the following methods is adopted for tunneling in soft soils? Pilot tunnel method Drift method Needle beam method Heading and benching method

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

359. [UNKNOWN] unknown

141. What is the minimum number of degrees of freedom that a robot needs to have in order to locate its end effectors at an arbitrary point with an arbitrary orientation in space? (a) 3 (b) 4 (c) 5 (d) 6

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

360. [UNKNOWN] unknown

142. Which one of the following features does not pertain to Littoral drift? It depends on length of wave It is the process of erosion or deposition by waves Waves caused by prevailing wind, stir up and move sand particles Wind tends to carry drifting sand in a zigzag way

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

361. [UNKNOWN] unknown

142. Using a robot with 1 degree of freedom and having 1 sliding joint with a full range of 1. m, if the robot's control memory has a 12-bit storage capacity, the control resolution for the axis of motion will be 0.236 mm 0.244 mm 0.252 mm 0.260 mm

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

362. [UNKNOWN] unknown

143. Consider the following data for designing a taxiway for operating Boeing 707 — 320 aeroplane : Wheel base = 17.70 m Tread of main landing gear = 6.62 m Turning speed = 40 km/h Coefficient of friction between tyres and pavement surface = 0.13 The turning radius of the taxiway will be 98.5 m 94.5 m 89.5 m 86.5 m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

363. [UNKNOWN] unknown

143. Assume that the joint mechanisms at serial link manipulators are frictionless. The joint torque T required to bear an arbitrary end point force F is, $r_1 F$ $J F$ $J^T F$ $J^{-1} F$

364. [UNKNOWN] unknown

144. Which one of the following instances of performance of aircraft is not considered for determining basic runway length? (a) Normal landing case Normal take-off case Engine failure case Emergency landing case VBY-S-CVL (23 — A:) ===== PAGE 24 ===== Codes: Both Statement (I) and Statement (II) are individually true, and Statement (II) is the correct explanation of Statement (I). Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I). Statement (I) is true, but Statement (II) is false. Statement (I) is false, but Statement (II) is true.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

365. [UNKNOWN] unknown

144. Rotate the vector $v = 5i + 3j + 8k$ by an angle of 90° about the x-axis. The rotated vector (Hu) would be (a) $(c) - 3 - 8 - 5 Po$ (d) $1 - 5 - 81 3 1$ VBY-S-MCH/50A 26 ===== PAGE 27 ===== Directions : Each of the six (6) items consists of two statements, one labelled as 'Statement (I)' and the other as 'Statement (II)'. You are to examine these two statements carefully and select the answers to these items using the code given below : Code: Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I) Both Statement (I) and Statement (H) are individually true but Statement (II) is not the correct explanation of Statement (I) Statement (I) is true but Statement (H) is false Statement (I) is false but Statement (H) is true Statement (I) : The function of arithmetic logic unit (ALU) in microprocessor is to perform data manipulation. Statement (II) : The status register is where data for an input to the arithmetic and logic unit is temporarily stored. Statement (I) : To use a sensor, we generally need to add signal conditioning circuitry, such as circuits which amplify and convert from analog to digital, to get the sensor signal in the right form, take account of any non-linearities, and calibrate it. Statement (II) : A smart sensor is integrated with the required buffering and conditioning circuitry in a single element and provides functions beyond that of just a sensor. Statement (I) : The count-up overflow (OV) bit is 1 when the up-counter increments above the maximum positive value. Statement (II) : The count-down underflow (UN) bit is 1 when the counter, decrements below the minimum negative value. Statement (I) : The multiplexer is essentially an electronic switching device which enables each of the inputs to be sampled in turn. Statement (II) : A multiplexer is a circuit that is able to have inputs of data from a number of sources and then, by selecting an input channel, gives an output from just one of them. Statement (I) : The term 'encoder' is used for a device that provides an analog output as a result of angular or linear displacement. Statement (II) : An increment encoder detects changes in angular or linear displacement from some datum position where as an absolute encoder gives the actual angular or linear position. Statement (I) : Process control valves are used to control the rate of fluid flow and are used where, perhaps, the rate of flow of a liquid into a tank has to be controlled. Statement (H) : A common form of pneumatic actuator used with process control valves is the diaphragm actuator. VBY-S-M CH /50A 27 [P.T.O.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/MECHANICAL-ENGINEERING-ESEP-19_0.pdf

366. [UNKNOWN] unknown

145. Statement (1): Expansive cement is used in repair work for opened up joints. Statement (II) : Expansive cement expands while hardening.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf

367. [UNKNOWN] unknown

146. Statement (I) : Plastic hinges are developed when stress at every point is equal to yield stress. Statement (II) : Plastic hinges are formed at sections subjected to the greatest curvature. If degree of fixity at supports is lessened, the maximum sagging moment at mid-span decreases. Statement U): Torsion reinforcement is provided at (and near) corners in a two-way slab which is simply supported on both edges meeting at the corner. Statement (II) : The area of reinforcement in each of the layers shall be three-quarters of the area required for maximum mid-span moment in the slab. Statement (I) : The inclination of the resultant stress with normal can exceed the angle of repose (adopting old terminology). Statement (II) : The ratio of the difference between greatest and least intensifies of pressure to their sum cannot exceed the sine of the angle of repose (adopting old terminology). Statement (I) : Alum works in slightly alkaline range. Statement (II) : At higher temperatures, viscosity of water (resistance to settling) decreases and flocs settle better. 147% Statement (I):. If degree of fixity at supports is lessened, the maximum hogging moment at the ends will decrease. Statement (II) : Directions : Each of the next six (06) items consists of two statements, one labeled as

'Statement (I)' and the other as 'Statement (H). You are to examine the two statements carefully and select the answers to these six items using the codes given below: VBY-S-CVL (24 — A)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/CIVIL-ENGINEERING-ESEP-19_0.pdf